

Compact and Tightly Secure (Anonymous) IBE from Module LWE in the QROM

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Abstract

We present a new compact and tightly secure (anonymous) identity-based encryption (IBE) scheme based on the module learning with errors (MLWE) assumption, known as the standard lattice assumption, in the (quantum) random oracle model. Our IBE scheme is the first to achieve the asymptotically most compact and tight security under the standard lattice assumption. We design our IBE scheme by instantiating the framework of Gentry, Peikert, and Vaikuntanathan (STOC'08) using the compact trapdoor proposed by Yu, Jia, and Wang (CRYPTO'23). The tightness of our IBE scheme is achieved by extending the proof technique of Katsumata et al. (ASIACRYPT'18, JoC'21) to the Hermite normal form setting. To achieve this, we develop some new results on module lattices that may be of independent interest.

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1 Introduction

1.1 Background

Identity-based encryption (IBE), introduced by Shamir [Sha84], is a generalization of public key encryption (PKE). Unlike traditional PKE, IBE allows senders to encrypt messages using a master public key mpk and an arbitrary string id , such as the recipient’s username or email address. This means IBEs do not require a Public Key Infrastructure (PKI). In addition, when communicating with multiple users, an IBE only needs one mpk , whereas a PKE requires as many public keys as there are users. Because of these advantages, IBE has been discussed in the context of several practical applications [AKG⁺07, BRTM08, DSDSAL08, TWZL08, ZC11, HSM13, MSW15]. Since the first IBE scheme was proposed in 2001 [BF01, Coc01], it has been improved in various ways [BGK08, Wat09, BKP14, DG17]. However, these traditional schemes are vulnerable to quantum attacks due to Shor’s algorithm [Sho99].

In 2008, Gentry, Peikert, and Vaikuntanathan [GPV08] proposed the first post-quantum IBE scheme (GPV-IBE) based on standard unstructured lattices. The GPV-IBE is secure under the learning with errors (LWE) assumption [Reg05] in the random oracle model (ROM). Since then, there have been several studies on lattice-based IBEs from different perspectives, including extending to the quantum ROM (QROM) [Zha12, KYY18, KYY21], removing the random oracle [ABB10a, CHKP10, KY16, BL16, Yam17, ALWW21], and adding security properties [ABB10b, AP12, NY19, KMT19, EKW19]. However, these constructions only indicate improvements on the theoretical side. In particular, these IBE schemes are still asymptotically inefficient even when instantiated on structured lattices [BFRS18, BEP⁺21]¹ such as the ML-KEM [SAB⁺22] and the ML-DSA [LDK⁺22]. On the practical side, Ducas, Lyubashevsky, and Prest [DLP14] proposed the first practical lattice-based IBE scheme (DLP-IBE) based on NTRU lattices, and then several works [MSO17, CKKS19, ZMS⁺24] optimized the DLP-IBE.

Unfortunately, even the DLP-IBE [DLP14] and its variants [MSO17, CKKS19, ZMS⁺24] have several efficiency challenges. One of these is the tightness of the security reduction. The efficiency of cryptographic schemes depends on the tightness of the security reduction. In general, we say that the security of a cryptographic scheme under a given computational assumption is tight if breaking the scheme’s security is as hard as solving the assumption. More precisely, suppose that we have proved that if there is an adversary who can break the security of the scheme with advantage ϵ and running time T , we can break the underlying assumption with advantage ϵ' and running time T' . We then obtain the inequality $\epsilon/T \leq L \cdot \epsilon'/T'$, where L is the reduction loss of the scheme. The scheme is tightly secure if $L = O(1)$. If the scheme is not tightly secure, we need to set the parameters larger to ensure the concrete security of the scheme. The DLP-IBE is not tightly secure because its security proof relies on the non-tight proof of the GPV-IBE [GPV08], and its reduction loss depends on the number of adversary queries. Several tightly secure lattice-based IBE schemes have also been proposed [BL16, BL18, LLW20, KYY18, KYY21, KTY23], but none is as efficient as the DLP-IBE. From the above, the natural question is:

Can we construct a compact and tightly secure IBE scheme from lattices?

1.2 Our Contributions

In this paper, we propose a new compact and tightly secure IBE scheme based on the module LWE (MLWE) assumption in the (Q)ROM. Our IBE scheme is the first to achieve the asymptotically most compact and tight security under the standard lattice assumption. As a result, we answer the above question in the affirmative. Furthermore, our IBE scheme satisfies anonymity by ensuring that the ciphertext does not reveal any information about the identity or the message.

In Table 1, we summarize our results and a comparison with previous lattice-based (anonymous) IBE schemes in the (Q)ROM. Table 1 shows that our scheme is asymptotically the most efficient among LWE-, RLWE-, and MLWE-based schemes when $k = O(1)$. The MLWE-based scheme allows flexible, broad

¹The schemes in these papers only provide selective security. Therefore, these schemes are not included in the comparison in Table 1.

parameter settings, including LWE and RLWE. In this sense, for the total dimension $N = nk$, our scheme is based on RLWE when $k = 1$ (i.e., $n = N$) and LWE when $k = N$ (i.e., $n = 1$). It should be noted that our scheme based on LWE (i.e., case of $k = N$) is more efficient than previous ones [GPV08, Zha12, KYY21], and ours based on RLWE (i.e., case of $k = 1$) is more efficient than previous ones [KYY21, JHTW24] as well. Furthermore, when assuming $k = O(1)$, our scheme is as compact as the one [DLP14] and has the advantage that the security loss is constant (i.e., tight).

Table 1: Comparison of adaptively-secure anonymous IBE schemes from lattices. $|\text{mpk}|$, $|\text{sk}|$, and $|\text{ct}|$ denote the size of the master public key, secret key, and ciphertext, respectively. $N = nk$ denotes the dimension of the underlying lattice, where n is the degree and k is the rank. For LWE, $n = 1$, and for RLWE and NTRU, $k = 1$. q denotes the modulus. $Q_H = \text{poly}(\lambda)$ and $Q_{\text{id}} = \text{poly}(\lambda)$ denote the number of (quantum) random oracle and secret key queries, respectively, where λ denotes the security parameter. $\nu = \nu(\lambda)$ denotes the advantage of the scheme. [†] Its security losses are based on the results of the previous work [GPV08] because the security proof is omitted in their paper.

Scheme	$ \text{mpk} $	$ \text{sk} , \text{ct} $	Assumption	Security loss
GPV [GPV08]	$O(N^2 \log^2 q)$	$O(N \log^2 q)$	LWE	$O(Q_H)$
Zhandry [Zha12]	$O(N^2 \log^2 q)$	$O(N \log^2 q)$	LWE	$O\left(\frac{(Q_H + Q_{\text{id}})^4}{\nu}\right)$
DLP [DLP14]	$O(N \log q)$	$O(N \log q)$	NTRU	$O(Q_H)$
KYY [KYY21, Sec. 3]	$O(N^2 \log^2 q)$	$O(N \log^2 q)$	LWE	$O(1)$
KYY [KYY21, Sec. 5.1]	$O(N \log^2 q)$	$O(N \log^2 q)$	RLWE	$O(1)$
HYCL [JHTW24] [†]	$O(N \log^2 q)$	$O(N \log^2 q)$	RLWE	$O(Q_H)$
Ours	$O(kN \log q)$	$O(N \log q)$	MLWE	$O(k)$

In addition to the aforementioned academic contributions, it is important from a practical standpoint to construct an efficient IBE scheme based on MLWE. This is because it shares the same assumptions as ML-KEM, the U.S. standard for post-quantum KEM/PKE, making it well-suited for real-world implementation.

Technical Overview. Here, we briefly summarize the spirit of our construction and security proof. Our proposed scheme is a GPV-IBE [GPV08] instantiated by Yu et al.’s compact preimage sampling [YJW23]. Hence, we first briefly describe the GPV-IBE.

GPV-IBE over module lattices. Let $\mathcal{R} = \mathbb{Z}_q[X]/(X^n + 1)$ and $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$. In the GPV-IBE, a master public key is a fat matrix $\mathbf{A} \in \mathcal{R}_q^{k \times \ell}$ and a master secret key is its trapdoor $\text{td}_{\mathbf{A}}$, which enables one to sample a short preimage $\mathbf{x} \in \mathcal{R}_q^\ell$ such that $\mathbf{A}\mathbf{x} = \mathbf{y} \bmod q$ given an arbitrary vector $\mathbf{y} \in \mathcal{R}_q^k$. A secret key sk_{id} for an identity $\text{id} \in \{0, 1\}^*$ is a short vector $\mathbf{x}_{\text{id}} \in \mathcal{R}^\ell$ such that $\mathbf{A}\mathbf{x}_{\text{id}} = \mathbf{y}_{\text{id}} \bmod q$, where $\mathbf{y}_{\text{id}} = \mathbf{H}(\text{id})$ for a hash function $\mathbf{H} : \{0, 1\}^* \rightarrow \mathcal{R}_q^k$. A ciphertext for a message $\mathbf{M} \in \{0, 1\}$ and an identity id consists of $\mathbf{c}_1 = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 \in \mathcal{R}_q^\ell$ and $c_2 = \mathbf{y}_{\text{id}}^\top \mathbf{r} + e_2 + \mathbf{M} \cdot \lfloor q/2 \rfloor \in \mathcal{R}_q$, where $\mathbf{r} \in \mathcal{R}_q^k$ is a uniform random vector, $\mathbf{e}_1 \in \mathcal{R}_q^\ell$ and $e_2 \in \mathcal{R}_q$ are small noise terms.

Tight proof by Katsumata et al. [KYY18, KYY21]. Katsumata et al. showed that GPV-IBE had tight security in the (Q)ROM. We outline the security proof in the ROM. To answer a random oracle query on id , the reduction algorithm chooses a random short vector $\mathbf{x}_{\text{id}} \in \mathcal{R}^\ell$ and sets $\mathbf{y}_{\text{id}} = \mathbf{A}\mathbf{x}_{\text{id}} \bmod q$. If $\mathbf{x}_{\text{id}}$ has sufficient entropy, $\mathbf{y}_{\text{id}}$ is uniformly distributed over $\mathcal{R}_q^k$. Using this fact, the reduction algorithm returns $\mathbf{y}_{\text{id}}$ for the random oracle query and $\mathbf{x}_{\text{id}}$ for the secret key query. Note that the reduction algorithm knows a secret key $\mathbf{x}_{\text{id}^*}$ for a target identity id^* . Thus, the reduction algorithm can simulate the challenge ciphertext by generating $\mathbf{c}_1^* = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1$ and $c_2^* = \mathbf{y}_{\text{id}^*}^\top \mathbf{r} + \mathbf{M} \cdot \lfloor q/2 \rfloor$. It is important to note that we no longer need the LWE instance $(\mathbf{y}_{\text{id}^*}, \mathbf{y}_{\text{id}^*}^\top \mathbf{r} + e_2)$ to simulate the challenge ciphertext. The actual proof uses the noise re-randomization technique by Katsumata and Yamada [KY16] to simulate the distribution of c_2^* (especially the noise term e_2).

More compact scheme via approximate preimage sampling. At the heart of the GPV-IBE is the preimage sampling technique, which is also a source of non-compactness. This is because the width of the matrix $\mathbf{A}$

must be $\ell = O(k \log_2 q)$ to realize preimage sampling. To improve the compactness, Chen et al. [CGM19] introduced the relaxed notion of preimage sampling, called *approximate preimage sampling*. With approximate preimage sampling, instead of sampling an exact preimage $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{y} \bmod q$, sample an approximate preimage $\mathbf{x} \in \mathcal{R}^\ell$ such that $\mathbf{Ax} = \mathbf{y} - \mathbf{z} \bmod q$, where $\mathbf{z} \in \mathcal{R}^k$ is a short error vector. Recently, Yu et al. [YJW23] developed a compact framework for approximate preimage sampling that uses a *nearly square matrix* instead of the short and fat one used in [CGM19].

To construct an efficient IBE scheme, we instantiate the GPV-IBE using Yu et al.’s approximate preimage sampling. Very recently and concurrently, Izabachène et al. [IPR23]² and Jia et al. [JHTW24] proposed compact IBE schemes by using the approximate preimage sampling. The design idea of our scheme is similar to their schemes. Namely, to encrypt a message $\mathbf{M}$ under an identity id , we use a short random vector $\mathbf{r} \in \mathcal{R}^k$ instead of a uniform random vector. This is to keep the error term $\mathbf{z}^\top \mathbf{r}$ that appears during decryption small, where $\mathbf{z}$ is an approximate error.

Attempt: Apply the Katsumata et al. security proof directly. We try to apply the proof techniques of Katsumata et al. to the above compact scheme. Mostly, their proof technique can be applied, but there is one part where it cannot. This is the part that simulates the c^* of the challenge ciphertext. In their proof, they use a secret key $\mathbf{x}_{\text{id}^*}$, which is an exact preimage vector, and the noise re-randomization technique of Katsumata and Yamada [KY16] to approximately simulate c^* . In our scheme, $\mathbf{x}_{\text{id}^*}$ is an approximate preimage vector rather than an exact preimage vector. Then, when we try to simulate c^* , we have

$$\begin{aligned} c_2^* &= \mathbf{x}_{\text{id}^*}^\top \mathbf{c}_1^* + \mathbf{M} \cdot \lfloor q/2 \rfloor \\ &= \mathbf{x}_{\text{id}^*}^\top (\mathbf{A}^\top \mathbf{r} + \mathbf{e}_1) + \mathbf{M} \cdot \lfloor q/2 \rfloor \\ &= (\mathbf{Ax}_{\text{id}^*})^\top \mathbf{r} + \mathbf{x}_{\text{id}^*}^\top \mathbf{e}_1 + \mathbf{M} \cdot \lfloor q/2 \rfloor \\ &= \mathbf{y}_{\text{id}^*}^\top \mathbf{r} - \boxed{\mathbf{z}^\top \mathbf{r}} + \mathbf{z}_{\text{id}^*}^\top \mathbf{e}_1 + \mathbf{M} \cdot \lfloor q/2 \rfloor. \end{aligned}$$

Unfortunately, the noise re-randomization technique cannot account for this additional error term $\mathbf{z}^\top \mathbf{r}$ that appears by evaluating $\mathbf{Ar}_{\text{id}^*}$. Therefore, we must use a different approach to complete the proof.

Solution: MLWE with an adaptive hint. To overcome the above problem, we use the MLWE with an adaptive hint about the MLWE secret and error, instead of the noise re-randomization technique. This assumption is a Hermite normal form of the MLWE with *error-only* leakage introduced by Döttling et al. [DKL⁺23]. Roughly speaking, this assumption says that the MLWE assumption holds even if a hint of the secret $\mathbf{r}$ and the noise $\mathbf{e}_1$ are adaptively given. We show a reduction from the (standard) MLWE problem to this variant. This allows us to exactly simulate c_2^* by using an approximate preimage vector $\mathbf{x}_{\text{id}^*}$ and a hint without the noise re-randomization. Therefore, we can complete the proof.

Finally, we note that the above proof naturally fits in the QROM setting similar to [KYY18, KYY21]. Thus, the proof in the classical ROM can be almost automatically converted into the one in the QROM. Our proof technique can also be seen as an extension of [KYY18, KYY21]’s proof technique to the Hermite normal form (HNF) setting. Furthermore, our proof technique is somewhat general since it can be applied to any approximate trapdoor sampling, e.g., [CGM19, JHTW24, JRS24], by appropriately setting parameters.

Comparison with the previous version in PQCrypto 2024. Here, we highlight the new contributions of the current paper, beyond the previous version [TS24] published in PQCrypto 2024. We summarize the comparison between this paper and its previous versions in Table 2.

Table 2: A comparison between this paper and the previous version [TS24].

	Assumption	Parameters
Previous version [TS24]	RLWE	Depending on # of the adversary’s queries
This paper	MLWE	Independent on # of the adversary’s queries

²Their proposed scheme is *selectively* secure in the *standard model*.

The main difference from the PQCrypto 2024 version is that it uses module lattices instead of ideal ones. This fundamental shift allows for a more flexible trade-off between efficiency and scalability than ideal lattices, addressing a key practical limitation. Consequently, this paper introduces drastically new technical results on module lattices, extending previous findings. To this end, we develop three new results on module lattices that may be of independent interest.

The first result is a tight reduction of the MLWE problem to a variant of MLWE where a hint about the secret and the noise is given *adaptively*, alongside a tighter security proof based on this variant. The PQCrypto 2024 version relied on the multi-hint extended RLWE problem introduced by Mera et al. [MKMS22]. Critically, while that problem could be extended to the module lattice setting, achieving adaptive security would necessitate numerous hints dependent on the adversary’s query count. This inherently led to an undesirable scenario where certain parameters were tied to the number of queries, undermining tight security. To resolve this significant challenge, we adopt the MLWE with *an adaptive hint* (ahMLWE) problem. This novel approach allows for security proofs with only one hint, enabling parameters to be set fully independently of the number of queries, thereby achieving true tightness.

The second insight is a new Gaussian regularity lemma over rings. Specifically, we provide a new Gaussian regularity *with leakage* over the ring with an *almost prime* modulus. Unlike the PQCrypto 2024 version, which, though capable of achieving tightness, employed a less optimized approach, our improved technique results in simpler security proofs and significantly smaller parameters, directly contributing to the scheme’s compactness.

The third result is a compact approximate trapdoor over module lattices. This is a simple extension of the approximate trapdoor of Yu et al. [YJW23]. For completeness and readability, we provide a detailed description of our approximate trapdoor and its correctness. These results are summarized in Section 3.

Organization

This paper is organized as follows. In Section 2, we first recall the notations, cryptographic definitions, and related lemmas. In Section 3, we show new results over module lattices. In Section 4, we present the description of our IBE scheme. In Section 5, we give a security proof of our IBE scheme in the ROM. In Section 6, we provide a security proof of our IBE scheme in the QROM. In Section 7, we finally conclude the paper.

2 Preliminaries

Notations. Let λ denote the security parameter throughout the paper. We denote by $[n]$ the set $\{1, \dots, n\}$ for any positive integer. For a finite set $\mathcal{S}$, let $\mathcal{U}(\mathcal{S})$ be the uniform distribution over $\mathcal{S}$ and let $s \leftarrow \mathcal{S}$ denote the operation of sampling a from $\mathcal{S}$ uniformly at random. For a probability distribution or random variable $\mathcal{X}$, let $x \leftarrow \mathcal{X}$ denote the operation of sampling x according to $\mathcal{X}$. Let $\mathcal{X}$ and $\mathcal{Y}$ be two random variables over some finite set $\mathcal{S}_X$ and $\mathcal{S}_Y$, respectively. The statistical distance $\Delta(\mathcal{X}, \mathcal{Y})$ between $\mathcal{X}$ and $\mathcal{Y}$ is defined as $\Delta(\mathcal{X}, \mathcal{Y}) := \frac{1}{2} \sum_{s \in \mathcal{S}_X \cup \mathcal{S}_Y} |\Pr[X = s] - \Pr[Y = s]|$. We say that $\mathcal{X}$ and $\mathcal{Y}$ are statistically close and denote as $\mathcal{X} \approx_s \mathcal{Y}$ when $\Delta(\mathcal{X}, \mathcal{Y}) = \text{negl}(\lambda)$. For two distributions $\mathcal{X}$ and $\mathcal{Y}$, we denote the convolution of $\mathcal{X}$ and $\mathcal{Y}$ by $\mathcal{X} * \mathcal{Y}$.

2.1 Linear Algebra, Lattices, and Gaussian

Linear Algebra. Any matrix $\mathbf{M} \in \mathbb{R}^{n \times m}$ can be written as $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ and $\mathbf{V} \in \mathbb{R}^{m \times m}$ are orthogonal matrices and $\mathbf{D} \in \mathbb{R}^{n \times m}$ is an upper diagonal matrix (singular value decomposition). The entries of $\mathbf{D}$ are called the singular values of $\mathbf{M}$, and we denote the smallest singular value by $\sigma_{\min}(\mathbf{M})$ and the largest singular value by $\sigma_{\max}(\mathbf{M})$. The largest singular value $\sigma_{\max}(\mathbf{M})$ is equal to the Euclidean spectral norm $\|\mathbf{M}\|_2 := \max_{\|\mathbf{x}\|_2=1} \|\mathbf{M}\mathbf{x}\|_2$. We will use the following bound for the largest singular value of a short matrix.

Lemma 2.1 ([Lan23, Lemma 1]). Let $n, m \in \mathbb{N}$, $\beta > 0$ be a positive real, and $\mathbf{M} \in \mathbb{R}^{n \times m}$ be a matrix such that $\|\mathbf{M}\|_\infty \leq \beta$. Then, it holds that $\sigma_{\max}(\mathbf{M}) \leq \beta\sqrt{n}$.

Lattices. A lattice $\Lambda \subseteq \mathbb{R}^n$ is the set of all integer-linear combinations of a set of linearly independent basis vectors, i.e., for any lattice Λ , there exists a full-rank matrix $\mathbf{B}^{n \times m}$ such that $\Lambda = \Lambda(\mathbf{B}) = \{\mathbf{B}\mathbf{z} \mid \mathbf{z} \in \mathbb{Z}^m\}$. We call m the rank of Λ and $\mathbf{B}$ a basis of Λ , and we say that Λ is full-rank if $m = n$. The dual of a lattice Λ is $\Lambda^* := \{\mathbf{w} \in \mathbb{R}^n \mid \forall \mathbf{v} \in \Lambda : \mathbf{v}^\top \mathbf{w} \in \mathbb{Z}\}$. Given a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ for some $n, m, q \in \mathbb{N}$, we can define the following q -ary lattices:

$$\begin{aligned}\Lambda_q(\mathbf{A}) &:= \{\mathbf{y} \in \mathbb{Z}^m : \mathbf{y} = \mathbf{A}^\top \mathbf{s} \bmod q \text{ for some } \mathbf{s} \in \mathbb{Z}^n\}, \\ \Lambda_q^\perp(\mathbf{A}) &:= \{\mathbf{y} \in \mathbb{Z}^m : \mathbf{A}\mathbf{y} = \mathbf{0} \bmod q\}.\end{aligned}$$

In this paper, we will deal with lattices of the form $\mathcal{R}^k$ and $\mathcal{R}_q^k$, where $\mathcal{R} := \mathbb{Z}[X]/(X^n + 1)$ and $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ for $q \in \mathbb{N}$ are rings. The ring $\mathcal{R}$ becomes a lattice through the coefficient embedding $\psi : \mathcal{R} \rightarrow \mathbb{Z}^n$ that maps every $a = \sum_{i=0}^{n-1} a_i X^i \in \mathcal{R}$ to its coefficient vector $\psi(a) = (a_0, \dots, a_{n-1})^\top \in \mathbb{Z}^n$. We extend ψ component-wise to vectors and matrices over $\mathcal{R}$. The embedding also induces a norm on the ring elements $\mathbf{a} \in \mathcal{R}^k$. That is, we define $\|\mathbf{a}\|_\infty := \|\psi(\mathbf{a})\|_\infty$. The multiplication in $\mathcal{R}$ translates into a matrix-vector multiplication once embedded with ψ . For all $a, b \in \mathcal{R}$, we can write $\psi(a \cdot b)$ as $\text{Rot}(a) \cdot \psi(b)$, where $\text{Rot}(a)$ is defined as

$$\text{Rot}(a) = (\psi(a), \psi(aX), \dots, \psi(aX^{n-1})) = \begin{pmatrix} a_0 & -a_{n-1} & \cdots & -a_1 \\ a_1 & a_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & -a_{n-1} \\ a_{n-1} & a_{n-2} & \cdots & a_0 \end{pmatrix} \in \mathbb{Z}^{n \times n},$$

which is itself a nega-circulant matrix. We also extend Rot component-wise to vectors and matrices over $\mathcal{R}$. Let $\mathbf{I}_{\mathbb{Z},k} \in \mathbb{Z}^{k \times k}$ and $\mathbf{I}_{\mathcal{R},k} \in \mathcal{R}^{k \times k}$ be identity matrices on $\mathbb{Z}$ and $\mathcal{R}$, respectively.

Gaussian. Let $\Sigma \in \mathbb{R}^{n \times n}$ be a positive definite matrix, we define the Gaussian function $\rho_{\sqrt{\Sigma}}(\mathbf{x}) := \exp(-\pi \mathbf{x}^\top \Sigma^{-1} \mathbf{x})$ for any $\mathbf{x} \in \mathbb{R}^n$. For a lattice $\Lambda \subseteq \mathbb{R}^n$, we define the discrete Gaussian distribution

$$\mathcal{D}_{\Lambda, \sqrt{\Sigma}}(\mathbf{x}) := \frac{\rho_{\sqrt{\Sigma}}(\mathbf{x})}{\rho_{\sqrt{\Sigma}}(\Lambda)}$$

for any $\mathbf{x} \in \mathbb{R}^n$, where $\rho_{\sqrt{\Sigma}}(\Lambda) := \sum_{\mathbf{y} \in \Lambda} \rho_{\sqrt{\Sigma}}(\mathbf{y})$. When $\Sigma = \sigma^2 \mathbf{I}_{\mathbb{Z},n}$ for a positive real $\sigma \in \mathbb{R}$, we use σ as subscript instead of $\sqrt{\Sigma}$. For a ring $\mathcal{R}$, we write $\mathcal{D}_{\mathcal{R}, \sqrt{\Sigma}}$ for the distribution that samples $z \in \mathcal{R}$ with probability $\rho_{\psi(\mathcal{R}), \sigma}(\psi(z))$. As coined by [MR07], we define the smoothing parameter of a lattice Λ , parameterized by $\epsilon > 0$, by

$$\eta_\epsilon(\Lambda) := \min \{s > 0 \mid \rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \epsilon\}.$$

We will use the following properties of discrete Gaussian distributions and the smoothing parameter.

Lemma 2.2 ([Lyu12]). It holds that

$$\Pr_{z \leftarrow \mathcal{D}_{\mathcal{R}, \sigma}} [\|z\|_\infty > \sqrt{\lambda} \sigma] = \text{negl}(\lambda).$$

Lemma 2.3 ([Pei08, Lemma 3.5]). Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice with basis $\mathbf{B}$, and let $\epsilon \in (0, 1)$. Then, it holds that

$$\eta_\epsilon(\Lambda) \leq \frac{\sqrt{\ln(2n(1 + 1/\epsilon))/\pi}}{\lambda_1^\infty(\Lambda^*)},$$

where $\lambda_1^\infty(\Lambda^*)$ is the minimum distance of Λ^* , in the ℓ_∞ norm

Lemma 2.4 ([MKMS21, Lemma 9]). Let $n \geq 4$ be a power of 2 such that $X^n + 1$ splits into n linear factors modulo prime p and $\mathcal{R}_p = \mathbb{Z}_p[X]/(X^n + 1)$. Then, it holds that

$$\lambda_1^\infty(\Lambda_p(\mathbf{I}_{\mathcal{R},k} \parallel \mathbf{A})) \geq \frac{1}{\sqrt{n}} \cdot p^{1 - \frac{k}{k+\ell} - \frac{\epsilon}{k}}$$

except for a fraction of at most $2^n/p^{\epsilon n}$ of all $\mathbf{A} \in \mathcal{R}_p^{k \times \ell}$.

Lemma 2.5 ([AGJ⁺24, Lemma 2.6]). Let $k, \ell, q \in \mathbb{N}$, and $\mathbf{A} \in \mathcal{R}_q^{k \times \ell}$ such that $\mathbf{A}\mathcal{R}_q^\ell = \mathcal{R}_q^k$. Then, let $\epsilon > 0$ be negligible in λ and $\Sigma \in \mathbb{R}^{n\ell \times n\ell}$ such that $\Sigma - \eta_\epsilon(\Lambda_q^\perp(\mathbf{A}))^2 \cdot \mathbf{I}_{\mathbb{Z},n\ell}$ is positive semi-definite. Then, it holds that

$$\{\mathbf{u} : \mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R},\sqrt{\Sigma}}^\ell, \mathbf{u} := \mathbf{A}\mathbf{x} \bmod q\} \approx_s \{\mathbf{u} : \mathbf{u} \leftarrow \mathcal{R}_q^k\}.$$

Lemma 2.6 ([DKL⁺23, Theorem 2], simplified by [Lan23]). Let $k \in \mathbb{N}$, $\beta > 0$ be a positive real, and $\epsilon > 0$ be negligible in λ . Let $\mathbf{z} \in \mathcal{R}^k$ be a vector such that $\|\mathbf{z}\|_\infty \leq \beta$. Further let $\sigma_0, \tau_0 \in \mathbb{R}$ and $s, t \geq 2\sqrt{2}$ be positive reals such that $\sigma_0 \geq \eta_\epsilon(\mathcal{R}^m)$, $\tau_0 \geq \eta_\epsilon(\mathcal{R})$, and

$$t\tau_0 \geq \frac{\sqrt{(s^2 + 1)(s^2 + 2)}}{s} \sigma_0 \beta.$$

Then, for $\sigma := \sqrt{(s^2 + 1)}\sigma_0$, $\tau := \sqrt{(t^2 + 1)}\tau_0$, and $\sigma^* := s/2 \cdot \sigma_0$, there exists an efficiently sampleable distribution $\mathcal{F}$ on $\mathcal{R}^m \times \mathcal{R}$ such that

$$\left\{ (\mathbf{r}_1, \mathbf{z}^\top \mathbf{r}_1 + r_2) : \begin{array}{l} \mathbf{r}_1 \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^k, \\ r_2 \leftarrow \mathcal{D}_{\mathcal{R},\tau} \end{array} \right\} \approx_s \left\{ (\mathbf{r} + \mathbf{f}_1, f_2) : \begin{array}{l} \mathbf{r} \leftarrow \mathcal{D}_{\mathcal{R},\sigma^*}^k, \\ (\mathbf{f}_1, f_2) \leftarrow \mathcal{F} \end{array} \right\}.$$

Approximate Trapdoor. Yu et al. [YJW23] proposed a compact approximate trapdoor for *integer* or *ideal* lattices. We recall their results.

Lemma 2.7 ([YJW23, Theorem 2]). Let $n, p, q, Q \in \mathbb{N}$ such that $Q = pq$. There exists a probabilistic polynomial time (PPT) algorithm $\text{AppSampPreZ}(\cdot, \cdot, \cdot, \cdot)$ satisfying the following: Let $\mathbf{A} \in \mathbb{Z}_Q^{n \times 3n}$ and $\mathbf{T} \in \mathbb{Z}^{3n \times n}$ be matrices such that $\mathbf{A}\mathbf{T} = p \cdot \mathbf{I}_{\mathbb{Z},n} \bmod Q$. $\text{ApproxZ} \cdot \mathbf{A}^{-1}(\cdot)$ denote $\text{AppSampPreZ}(\mathbf{A}, \mathbf{T}, \cdot, \sigma_1)$. Then, it holds that

$$\left\{ (\mathbf{A}, \mathbf{x}, \mathbf{y}, \mathbf{z}) : \begin{array}{l} \mathbf{y} \leftarrow \mathbb{Z}_Q^n, \\ \mathbf{x} \leftarrow \text{ApproxZ} \cdot \mathbf{A}^{-1}(\mathbf{y}), \\ \mathbf{z} := \mathbf{y} - \mathbf{A}\mathbf{x} \bmod Q \end{array} \right\} \approx_s \left\{ (\mathbf{A}, \mathbf{x}, \mathbf{y}, \mathbf{z}) : \begin{array}{l} \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z},\sigma_1}^{3n}, \\ \mathbf{z} \leftarrow \mathbb{Z}_p^n, \\ \mathbf{y} := \mathbf{A}\mathbf{x} + \mathbf{z} \bmod Q \end{array} \right\}$$

for any $\sigma_1^2 \geq (q^2 + 1) \cdot (\sigma_{\max}(\mathbf{T})^2 + 1) \cdot \eta_\epsilon(\mathbb{Z}^n)$.

Module Learning with Errors. We recall the module learning with errors (MLWE) assumption.

Definition 2.8 (Module Learning with Errors (MLWE) [LS15]). Let $k, \ell, q \in \mathbb{N}$ and χ be an error distribution on $\mathcal{R}$. We say that the module learning with errors (MLWE) problem $\text{MLWE}_{k,\ell,q,\chi}$ is hard if for any PPT algorithm $\mathcal{A}$, it holds that

$$\text{Adv}_{k,\ell,q,\chi}^{\text{MLWE}}(\lambda, \mathcal{A}) := |\Pr[\mathcal{A}(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e}) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{u}) = 1]| = \text{negl}(\lambda),$$

where $\mathbf{A} \leftarrow \mathcal{R}_q^{k \times \ell}$, $\mathbf{s} \leftarrow \mathcal{R}_q^\ell$, $\mathbf{e} \leftarrow \chi^k$, and $\mathbf{u} \leftarrow \mathcal{R}_q^k$. We write $\text{MLWE}_{k,\ell,q,\sigma}$ as a shorthand for $\text{MLWE}_{k,\ell,q,\chi}$ when $\chi = \mathcal{D}_{\mathcal{R},\sigma}$.

Lemma 2.9 (Hardness of MLWE [LS15]). For any integers k, ℓ , and q and real σ such that $q = \text{poly}(\ell n)$, $k = \text{poly}(\ell)$, and $\sigma = \sqrt{\ell} \cdot \omega(\sqrt{\log n})$, the $\text{MLWE}_{k,\ell,q,\sigma}$ problem is as hard as the worst-case lattice generalized-independent-vector-problem in dimension $N = k\ell$ with approximation factor $\sqrt{8N\ell} \cdot \omega(\sqrt{\log n}) \cdot q/\sigma$.

2.2 Identity-Based Encryption

Here, we review the definition of identity-based encryption (IBE) by following [KYY21]. **Syntax.** An IBE scheme Π consists of the following four PPT algorithms.

- $\text{Setup}(1^\lambda) \rightarrow (\text{msk}, \text{mpk})$: The setup algorithm takes the security parameter λ as input and outputs a master secret key msk and a master public key mpk . It is assumed that the descriptions of the message space $\mathcal{M}$ and the identity space $\mathcal{ID}$ are implicitly included in mpk .
- $\text{KGen}(\text{msk}, \text{mpk}, \text{id}) \rightarrow \text{sk}_{\text{id}}$: The key-generation algorithm takes the master secret key msk , the master public key mpk , and an identity $\text{id} \in \mathcal{ID}$ as input, and outputs a secret key sk_{id} . It is assumed that id is implicitly included in sk_{id} .
- $\text{Enc}(\text{mpk}, \text{id}, M) \rightarrow \text{ct}$: The encryption algorithm takes the master public key mpk , an identity $\text{id} \in \mathcal{ID}$, and a message $M \in \mathcal{M}$, and outputs a ciphertext ct .
- $\text{Dec}(\text{sk}_{\text{id}}, \text{ct}) \rightarrow M$: The decryption algorithm takes a secret key sk_{id} and a ciphertext ct and outputs a message M .

Correctness. We require that for all $\lambda \in \mathbb{N}$, $\text{id} \in \mathcal{ID}$, and $M \in \mathcal{M}$, it holds that

$$\Pr \left[\begin{array}{l} (\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda) \\ \text{Dec}(\text{sk}_{\text{id}}, \text{ct}) = M : \begin{array}{l} \text{sk}_{\text{id}} \leftarrow \text{KGen}(\text{msk}, \text{mpk}, \text{id}) \\ \text{ct} \leftarrow \text{Enc}(\text{mpk}, \text{id}, M) \end{array} \end{array} \right] = 1 - \text{negl}(\lambda).$$

Security. Let Π be an IBE scheme. The adaptive-identity anonymity is defined via a game between an adversary $\mathcal{A}$ and the challenger $\mathcal{C}$.

1. Setup Phase: $\mathcal{C}$ first runs $(\text{msk}, \text{mpk}) \leftarrow \text{Setup}(1^\lambda)$ and gives mpk to $\mathcal{A}$. It then prepares a set $\mathcal{Q}_{\text{sk}} := \emptyset$.
2. Query Phase: $\mathcal{A}$ may adaptively make the following two types of queries to $\mathcal{C}$:

Key generation query: Upon a query $\text{id} \in \mathcal{ID}$ from $\mathcal{A}$, $\mathcal{C}$ checks if $(\text{id}, \text{sk}_{\text{id}}) \in \mathcal{Q}_{\text{sk}}$, and returns sk_{id} to $\mathcal{A}$ if this is the case. Otherwise, $\mathcal{C}$ computes $\text{sk}_{\text{id}} \leftarrow \text{KGen}(\text{msk}, \text{mpk}, \text{id})$, stores $(\text{id}, \text{sk}_{\text{id}})$ in $\mathcal{Q}_{\text{sk}}$, and returns it to $\mathcal{A}$.

Challenge query: $\mathcal{A}$ is allowed to make this query only once. Upon a query $(\text{id}^*, M^*) \in \mathcal{ID} \times \mathcal{M}$ from $\mathcal{A}$, $\mathcal{C}$ checks if $(\text{id}^*, *) \notin \mathcal{Q}_{\text{sk}}$, and returns $\perp$ to $\mathcal{A}$ if this is not the case. Otherwise, $\mathcal{C}$ stores $(\text{id}^*, \perp)$ in $\mathcal{Q}_{\text{sk}}$ and chooses $\text{coin} \leftarrow \{0, 1\}$. If $\text{coin} = 0$, it runs $\text{ct}^* \leftarrow \text{Enc}(\text{mpk}, \text{id}^*, M^*)$. Otherwise, it randomly samples ct^* from a ciphertext space. Finally, $\mathcal{C}$ returns ct^* to $\mathcal{A}$.

3. Guess Phase: At some point, $\mathcal{A}$ outputs a guess $\widehat{\text{coin}} \in \{0, 1\}$ for coin and terminates.

The above completes the description of the game. In this game, the advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_{\Pi}^{\text{IBE}}(\lambda, \mathcal{A}) := |\Pr[\widehat{\text{coin}} = \text{coin}] - 1/2|.$$

We say that an IBE scheme Π satisfies *adaptive-identity anonymity* if the advantage $\text{Adv}_{\Pi}^{\text{IBE}}(\lambda, \mathcal{A})$ is negligible for all PPT adversaries $\mathcal{A}$.

3 New Results on Module Lattices

In this section, we present our new results on module lattices, which are employed in the security proof of our IBE scheme and may be of independent interest.

3.1 Module-LWE with an Adaptive Hint

Here, we introduce a variant of the MLWE assumption which allows an adversary to *adaptively* learn both the leakages of the MLWE *secret and error*. This assumption is a Hermite normal form of the MLWE with *error-only* leakage introduced in [DKL⁺23].

Definition 3.1 (MLWE with an Adaptive Hint (ahMLWE)). Let $k, \ell, q \in \mathbb{N}$, $\beta > 0$ be a positive real, and χ and χ' be error distributions on $\mathcal{R}$. The MLWE with adaptive hint (ahMLWE) problem $\text{ahMLWE}_{k,\ell,q,\chi,\chi',\beta}$ is defined via the following experiment, where $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ is a two-stage PPT algorithm.

1. The challenger $\mathcal{C}$ samples $\mathbf{A} \leftarrow \mathcal{R}_q^{k \times \ell}$ and provides $\mathbf{A}$ to $\mathcal{A}_1$.
2. $\mathcal{A}_1$ sends $\mathbf{z} \in \mathcal{R}^{k+\ell}$ such that $\|\mathbf{z}\|_\infty \leq \beta$ to $\mathcal{C}$.
3. $\mathcal{C}$ samples $\mathbf{s} \leftarrow \chi^\ell$, $\mathbf{e} \leftarrow \chi^k$, and $e' \leftarrow \chi'$ and sets $h := \mathbf{z}^\top \begin{pmatrix} \mathbf{s} \\ \mathbf{e} \end{pmatrix} + e' \bmod q$.
4. $\mathcal{C}$ chooses a random bit $b \leftarrow \{0, 1\}$.
5. If $b = 0$, $\mathcal{C}$ sets $\mathbf{u} := \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q$, otherwise, $\mathcal{C}$ samples $\mathbf{u} \leftarrow \mathcal{R}_q^k$.
6. $\mathcal{C}$ now runs $\mathcal{A}_2$ on input $(\mathbf{A}, \mathbf{u}, h)$, upon which $\mathcal{A}_2$ outputs a bit b' .

We say that $\text{ahMLWE}_{k,\ell,q,\chi,\chi',\beta}$ is hard if every PPT algorithm $\mathcal{A}$, it holds that

$$\text{Adv}_{k,\ell,q,\chi,\chi',\beta}^{\text{ahMLWE}}(\lambda, \mathcal{A}) := \Pr[b = b'] = \frac{1}{2} + \text{negl}(\lambda)$$

in the above experiment. We may write $\text{ahMLWE}_{k,\ell,q,\sigma,\sigma',\beta}$ as a shorthand for $\text{ahMLWE}_{k,\ell,q,\chi,\chi',\beta}$ when $\chi = \mathcal{D}_{\mathcal{R},\sigma}$ and $\chi' = \mathcal{D}_{\mathcal{R},\sigma'}$.

We prove that the standard MLWE implies the ahMLWE tightly with only a small parameter loss for suitable discrete Gaussian distributions.

Theorem 3.2 (Hardness of ahMLWE). Let $\beta > 0$ be a parameter and $\epsilon > 0$ be negligible in λ . Let $\sigma_0, \sigma, \sigma^*, \tau_0, \tau \in \mathbb{R}$ and $s, t \geq 2\sqrt{2}$ be positive reals such that as in the statement in Theorem 2.6. Then, assuming that $\text{MLWE}_{k,\ell,q,\sigma^*}$ is hard, $\text{ahMLWE}_{k,\ell,q,\sigma,\tau,\beta}$ is also hard.

More precisely, for any PPT algorithm $\mathcal{A}$, there exists a PPT algorithm $\mathcal{B}$ such that

$$\text{Adv}_{k,\ell,q,\sigma,\tau,\beta}^{\text{ahMLWE}}(\lambda, \mathcal{A}) = \text{Adv}_{k,\ell,q,\sigma^*}^{\text{MLWE}}(\lambda, \mathcal{B}) + \text{negl}(\lambda).$$

Proof. The proof is almost identical to that of [DKL⁺23, Theorem 3]. Let $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ be a PPT algorithm against the ahMLWE assumption and $\mathcal{F}$ be a distribution on $\mathcal{R}^{\ell+k} \times \mathcal{R}$ as in the statement in Theorem 2.6. We will construct the algorithm $\mathcal{B}$ as follows:

- Given an MLWE sample $(\mathbf{A}, \mathbf{u})$, provide $\mathbf{A}$ to $\mathcal{A}_1$, which outputs vectors $\mathbf{z}_0$ and $\mathbf{z}_1$.
- Sample $(\mathbf{f}_1, f_2) \leftarrow \mathcal{F}$.
- Compute $\mathbf{u}' := \mathbf{u} + (\mathbf{A} \parallel \mathbf{I}_{\mathcal{R},k})\mathbf{f}_1$ and $h := f_2$.
- Run $\mathcal{A}_2$ on input $(\mathbf{A}, \mathbf{u}', h)$ and output whatever $\mathcal{A}_2$ outputs.

If $(\mathbf{A}, \mathbf{u})$ is a well-formed $\text{MLWE}_{k,\ell,q,\sigma^*}$ sample, it holds that $\mathbf{u} = \mathbf{A}\mathbf{s} + \mathbf{e}$, where $\mathbf{s} \leftarrow \mathcal{D}_{\mathcal{R},\sigma^*}^\ell$ and $\mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R},\sigma^*}^k$. Consequently, by Theorem 2.6, it holds that

$$(\mathbf{A}, \mathbf{u}', h) \equiv \left(\mathbf{A}, (\mathbf{A} \parallel \mathbf{I}_{\mathcal{R},k}) \left(\begin{pmatrix} \mathbf{s} \\ \mathbf{e} \end{pmatrix} + \mathbf{f}_1 \right), f_2 \right)$$

$$\approx_s (\mathbf{A}, (\mathbf{A} \parallel \mathbf{I}_{\mathcal{R},k}) \mathbf{r}_1, \mathbf{z}^\top \mathbf{r}_1 + r_2),$$

where $\mathbf{r}_1 \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma}^{k+\ell}$ and $\mathbf{r}_2 \leftarrow \$ \mathcal{D}_{\mathcal{R},\tau}$. In this case, the sample computed by $\mathcal{B}$ is statistically close to a sample of $\text{ahMLWE}_{k,\ell,q,\sigma,\tau,\beta}$ for $b = 0$.

On the other hands, if $\mathbf{u}$ is distributed uniformly random, we can write $\mathbf{u}$ as $\mathbf{u} = \mathbf{y} + \mathbf{A}\mathbf{s} + \mathbf{e}$ for a uniform random $\mathbf{y} \leftarrow \$ \mathcal{R}_q^k$, $\mathbf{s} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma^*}^\ell$, and $\mathbf{e} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma^*}^k$. Consequently, in this case, it holds also by Theorem 2.6 that

$$\begin{aligned} (\mathbf{A}, \mathbf{u}', \mathbf{h}) &\equiv \left(\mathbf{A}, \mathbf{y} + (\mathbf{A} \parallel \mathbf{I}_{\mathcal{R},k}) \begin{pmatrix} \mathbf{s} \\ \mathbf{e} \end{pmatrix} + \mathbf{f}_1, f_2 \right) \\ &\approx_s (\mathbf{A}, \mathbf{y} + (\mathbf{A} \parallel \mathbf{I}_{\mathcal{R},k}) \mathbf{r}_1, \mathbf{z}^\top \mathbf{r}_1 + r_2) \\ &\equiv (\mathbf{A}, \mathbf{y}', \mathbf{z}^\top \mathbf{r}_1 + r_2), \end{aligned}$$

where $\mathbf{y}' \in \mathcal{R}_q^k$ is a uniformly random vector. In this case, the sample computed by $\mathcal{B}$ is statistically close to a sample of $\text{ahMLWE}_{k,\ell,q,\sigma,\tau,\beta}$ for $b = 1$. Putting these two facts together, we have

$$\text{Adv}_{k,\ell,q,\sigma,\tau,\beta}^{\text{ahMLWE}}(\lambda, \mathcal{A}) = \text{Adv}_{k,\ell,q,\sigma^*}^{\text{MLWE}}(\lambda, \mathcal{B}) + \text{negl}(\lambda).$$

□

3.2 Gaussian Regularity with Leakage

Here, we provide a new Gaussian regularity with leakage over $\mathcal{R}_Q$, where Q is not a prime but *almost prime*. Our result is a generalization of previous results in [SS11, SS13, MKMS21, MKMS22] that can be applied to over rings with almost prime modulus.

Theorem 3.3 (Gaussian Regularity with leakage). Let $n, k, \ell, p, q, Q = pq \in \mathbb{N}$ such that $n \geq 4$ is a power of 2 and $X^n + 1$ splits into n linear factors modulo prime p , $\epsilon > 0$ be negligible in λ , and $\sigma, \tau \in \mathbb{R}$ be a positive real such that $\sigma, \tau \geq q\sqrt{n \ln(2n(k+1+\ell)(1+1/\epsilon))/\pi} \cdot p^{\frac{k+1}{k+1+\ell} + \frac{\epsilon}{k+1}}$. Then, it holds that

$$\left\{ (\mathbf{A}, \mathbf{c}, \mathbf{y}, c) : \begin{array}{l} \mathbf{x} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma}^\ell, \\ \mathbf{z} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma}^k, \\ e \leftarrow \$ \mathcal{D}_{\mathcal{R},\tau}, \\ \mathbf{y} := \mathbf{A}\mathbf{x} + \mathbf{z} \bmod Q, \\ \boxed{c := \mathbf{c}^\top \mathbf{x} + e \bmod Q} \end{array} \right\} \approx_s \left\{ (\mathbf{A}, \mathbf{c}, \mathbf{y}, c) : \begin{array}{l} \mathbf{x} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma}^\ell, \\ \mathbf{z} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma}^k, \\ \mathbf{y} := \mathbf{A}\mathbf{x} + \mathbf{z} \bmod Q, \\ \boxed{c \leftarrow \$ \mathcal{R}_Q} \end{array} \right\},$$

where $\mathbf{A} \leftarrow \$ \mathcal{R}_Q^{k \times \ell}$ and $\mathbf{c} \leftarrow \$ \mathcal{R}_Q^\ell$.

Proof. By Theorem 3.4 and the parameter condition, we have

$$\begin{aligned} \sigma, \tau &\geq p\sqrt{n \ln(2n(k+1+\ell)(1+1/\epsilon))/\pi} \cdot q^{\frac{k+1}{k+1+\ell} + \frac{\epsilon}{k+1}} \\ &\geq \max \left\{ \eta_\epsilon \left(\Lambda_Q^\perp \left(\mathbf{I}_{\mathcal{R},k+1} \begin{pmatrix} \mathbf{A} \\ \mathbf{c}^\top \end{pmatrix} \right) \right), \eta_\epsilon(\Lambda_Q^\perp(\mathbf{I}_{\mathcal{R},k} \parallel \mathbf{A})) \right\}. \end{aligned}$$

This implies that the following matrices are positive semi-definite:

$$\begin{aligned} \begin{pmatrix} \sigma^2 \cdot \mathbf{I}_{\mathbb{Z},nk} & \mathbf{0} \\ \mathbf{0} & \tau^2 \end{pmatrix} - \eta_\epsilon \left(\Lambda_Q^\perp \left(\mathbf{I}_{\mathcal{R},k+1} \begin{pmatrix} \mathbf{A} \\ \mathbf{c}^\top \end{pmatrix} \right) \right)^2 \cdot \mathbf{I}_{\mathbb{Z},nk+1} &\in \mathbb{R}^{(nk+1) \times (nk+1)}, \\ (\sigma^2 - \eta_\epsilon(\Lambda_Q^\perp(\mathbf{I}_{\mathcal{R},k+1} \parallel \mathbf{A}))^2) \cdot \mathbf{I}_{\mathbb{Z},nk} &\in \mathbb{R}^{nk \times nk}. \end{aligned}$$

Therefore, by using Theorem 2.5 twice, we have

$$\left\{ (\mathbf{A}, \mathbf{c}, \mathbf{y}, c) : \begin{array}{l} \mathbf{x} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma}^\ell, \\ \mathbf{z} \leftarrow \$ \mathcal{D}_{\mathcal{R},\sigma}^k, \\ e \leftarrow \$ \mathcal{D}_{\mathcal{R},\tau}, \\ \mathbf{y} := \mathbf{A}\mathbf{x} + \mathbf{z} \bmod Q, \\ c := \mathbf{c}^\top \mathbf{x} + e \bmod Q \end{array} \right\} \approx_s \left\{ (\mathbf{A}, \mathbf{c}, \mathbf{y}, c) : \begin{array}{l} \boxed{\mathbf{y} \leftarrow \$ \mathcal{R}_Q^k}, \\ \boxed{c \leftarrow \$ \mathcal{R}_Q} \end{array} \right\}$$

$$\approx_s \left\{ (\mathbf{A}, \mathbf{c}, \mathbf{y}, c) : \begin{array}{l} \mathbf{x} \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^\ell, \\ \mathbf{z} \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^k, \\ \boxed{\mathbf{y} := \mathbf{A}\mathbf{x} + \mathbf{z} \bmod Q}, \\ c \leftarrow \$ \mathcal{R}_Q \end{array} \right\}.$$

□

We show the following lemma to complete the proof of Theorem 3.3.

Lemma 3.4. Let $n, k, \ell, p, q \in \mathbb{N}$ such that $n \geq 4$ is a power of 2 and $X^n + 1$ splits into n linear factors modulo prime p , and $\epsilon \in (0, 1)$ be a positive real. Then, it holds that

$$\eta_\epsilon(\Lambda_{pq}^\perp(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A})) \leq q \cdot \sqrt{n \ln(2n(k + \ell)(1 + 1/\epsilon))/\pi} \cdot p^{\frac{k}{k+\ell} + \frac{\epsilon}{k}}.$$

with all but negligible probability, where $\mathbf{A} \leftarrow \$ \mathcal{R}_{pq}^{k \times \ell}$.

Proof. By Theorem 2.3, we have

$$\eta_\epsilon(\Lambda_{pq}^\perp(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A})) \leq \frac{\sqrt{\ln(2n(k + \ell)(1 + 1/\epsilon))/\pi}}{\lambda_1^\infty(\Lambda_{pq}^\perp(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A})^*)} = \frac{\sqrt{\ln(2n(k + \ell)(1 + 1/\epsilon))/\pi}}{\frac{1}{pq} \cdot \lambda_1^\infty(\Lambda_{pq}(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A}))}.$$

Since $\Lambda_{pq}(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A}) \subseteq \Lambda_p(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A})$, we have $\lambda_1^\infty(\Lambda_{pq}(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A})) \geq \lambda_1^\infty(\Lambda_p(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A}))$. By Theorem 2.4, we have

$$\lambda_1^\infty(\Lambda_p(\mathbf{I}_{\mathcal{R}, k} \| \mathbf{A})) \geq \frac{1}{\sqrt{n}} \cdot p^{1 - \frac{k}{k+\ell} - \frac{\epsilon}{k}}$$

for a uniformly random matrix $\mathbf{A}$ with all but negligible probability. Combining the above completes our proof of the lemma. □

3.3 Compact Approximate Trapdoor for Module Lattices

Yu et al. [YJW23] proposed a compact approximate trapdoor for *integer* or *ideal* lattices. We provide a simple extension of their results to the *module* lattice setting.

Theorem 3.5 (Compact Approximate Trapdoor for Module Lattices). Let $k, n, p, q, Q \in \mathbb{N}$ such that $k = O(1)$ and $Q = pq$. There exist PPT algorithms (**AppTrapGen**, **AppSampPre**) satisfying the following:

- **AppTrapGen**($1^k, p, q, \sigma_0$) takes as input positive integers $k, p, q \in \mathbb{N}$ and a positive real $\sigma_0 > 0$, and returns a matrix-approximate trapdoor pair $(\mathbf{A}, \mathbf{T}_\mathbf{A}) \in \mathcal{R}_Q^{k \times 2k} \times \mathcal{R}_Q^{2k \times k}$, where $Q = pq$.
- Let $\mathbf{A}$ be generated by **AppTrapGen**, **Approx.** $\mathbf{A}^{-1}(\cdot)$ denote the approximate preimage sampling algorithm, **AppSampPre**($\mathbf{A}, \mathbf{T}_\mathbf{A}, \cdot, \sigma_1$). The following two distributions are statistically close:

$$\left\{ (\mathbf{A}, \mathbf{x}, \mathbf{y}, \mathbf{z}) : \begin{array}{l} \mathbf{y} \leftarrow \$ \mathcal{R}_Q^k, \\ \mathbf{x} \leftarrow \$ \text{Approx.}\mathbf{A}^{-1}(\mathbf{y}), \\ \mathbf{z} := \mathbf{y} - \mathbf{A}\mathbf{x} \bmod Q \end{array} \right\} \approx_s \left\{ (\mathbf{A}, \mathbf{x}, \mathbf{y}, \mathbf{z}) : \begin{array}{l} \mathbf{x} \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma_1}^{2k}, \\ \mathbf{z} \leftarrow \$ (\mathcal{D}_{\mathcal{R}, \sigma_1} * \mathcal{R}_p)^k, \\ \mathbf{y} := \mathbf{A}\mathbf{x} + \mathbf{z} \bmod Q \end{array} \right\}$$

for any $\sigma_1^2 \geq (q^2 + 1) \cdot (3kn\lambda\sigma_0^2 + 1) \cdot \eta_\epsilon(\mathcal{R}^k)$.

- The matrix $\mathbf{A}$ generated by **AppTrapGen** is computationally indistinguishable from a uniform random matrix on $\mathcal{R}_Q^{k \times 2k}$ assuming $\text{MLWE}_{k, k, Q, \sigma_0}$ assumption.

Proof. We first describe (**AppTrapGen**, **AppSampPre**).

- **AppTrapGen**($1^k, p, q, \sigma_0$) $\rightarrow (\mathbf{A}, \mathbf{T}_\mathbf{A}) \in \mathcal{R}_Q^{k \times 2k} \times \mathcal{R}_Q^{2k \times k}$:

1. Sample $\bar{\mathbf{A}} \leftarrow \mathcal{R}_Q^{k \times k}$, $\mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}, \sigma_0}^{k \times k}$, and $\mathbf{E} \leftarrow \mathcal{D}_{\mathcal{R}, \sigma_0}^{k \times k}$.
2. Compute

$$\mathbf{A} := (\bar{\mathbf{A}} \| p \cdot \mathbf{I}_{\mathcal{R}, k} + \bar{\mathbf{A}} \mathbf{S} + \mathbf{E}) \bmod Q \in \mathcal{R}_Q^{k \times 2k}, \quad \mathbf{T}_{\mathbf{A}} := \begin{pmatrix} -\mathbf{E} \\ -\mathbf{S} \end{pmatrix} \in \mathcal{R}_Q^{2k \times k}.$$

3. Output $(\mathbf{A}, \mathbf{T}_{\mathbf{A}})$.
- AppSampPre $(\mathbf{A}, \mathbf{T}_{\mathbf{A}}, \mathbf{y}, \sigma_1) \rightarrow \mathbf{x} \in \mathcal{R}^{2k}$:
 1. Set

$$\mathbf{A}_{\mathbb{Z}} := (\mathbf{I}_{\mathbb{Z}, kn} \| \text{Rot}(\mathbf{A})) \in \mathbb{Z}_Q^{nk \times 3nk}, \quad \mathbf{T}_{\mathbb{Z}} := \begin{pmatrix} \text{Rot}(\mathbf{T}_{\mathbf{A}}) \\ \mathbf{I}_{\mathbb{Z}, nk} \end{pmatrix} \in \mathbb{Z}_Q^{3nk \times nk}.$$

2. Sample $\mathbf{x}_{\mathbb{Z}} \leftarrow \text{AppSampPre}_{\mathbb{Z}}(\mathbf{A}_{\mathbb{Z}}, \mathbf{T}_{\mathbb{Z}}, \psi(\mathbf{y}), \sigma_1)$.
3. Parse $\mathbf{x}_{\mathbb{Z}} = \begin{pmatrix} \bar{\mathbf{x}}_{\mathbb{Z}} \\ \underline{\mathbf{x}}_{\mathbb{Z}} \end{pmatrix} \in \mathbb{Z}^{3nk}$, where $\bar{\mathbf{x}}_{\mathbb{Z}} \in \mathbb{Z}^{kn}$ and $\underline{\mathbf{x}}_{\mathbb{Z}} \in \mathbb{Z}^{2nk}$.
4. Output $\mathbf{x} := \psi^{-1}(\underline{\mathbf{x}}_{\mathbb{Z}}) \in \mathcal{R}^{2k}$.

Then, we show that the algorithm $\text{Approx}.\mathbf{A}^{-1}(\cdot) = \text{AppSampPre}(\mathbf{A}, \mathbf{T}_{\mathbf{A}}, \cdot, \sigma_1)$ correctly works. From the description, we have

$$\begin{aligned} \mathbf{A}_{\mathbb{Z}} \mathbf{T}_{\mathbb{Z}} &= (\mathbf{I}_{\mathbb{Z}, nk} \| \text{Rot}(\mathbf{A})) \cdot \begin{pmatrix} \text{Rot}(\mathbf{T}_{\mathbf{A}}) \\ \mathbf{I}_{\mathbb{Z}, nk} \end{pmatrix} \\ &= (\mathbf{I}_{\mathbb{Z}, nk} \| \text{Rot}(\bar{\mathbf{A}}) \| p \cdot \mathbf{I}_{\mathbb{Z}, nk} + \text{Rot}(\bar{\mathbf{A}} \mathbf{S} + \mathbf{E})) \cdot \begin{pmatrix} -\text{Rot}(\mathbf{E}) \\ -\text{Rot}(\mathbf{S}) \\ \mathbf{I}_{\mathbb{Z}, nk} \end{pmatrix} \\ &= -\text{Rot}(\mathbf{E}) - \text{Rot}(\bar{\mathbf{A}} \mathbf{S}) + p \cdot \mathbf{I}_{\mathbb{Z}, nk} + \text{Rot}(\bar{\mathbf{A}} \mathbf{S} + \mathbf{E}) \\ &= p \cdot \mathbf{I}_{\mathbb{Z}, nk}. \end{aligned}$$

Furthermore, by Theorems 2.1 and 2.2, we have

$$\sigma_1^2 \geq (q^2 + 1) \cdot (3kn\lambda\sigma_0^2 + 1) \cdot \eta_{\epsilon}(\mathcal{R}^k) \geq (q^2 + 1) \cdot (\sigma_{\max}(\mathbf{T}_{\mathbb{Z}})^2 + 1) \cdot \eta_{\epsilon}(\mathcal{R}^k)$$

with overwhelming probability. Thus, AppSampPre $_{\mathbb{Z}}$ correctly works, and we have

$$\mathbf{A}' \mathbf{x}' = (\mathbf{I}_{\mathbb{Z}, kn} \| \text{Rot}(\mathbf{A})) \begin{pmatrix} \bar{\mathbf{x}}_{\mathbb{Z}} \\ \underline{\mathbf{x}}_{\mathbb{Z}} \end{pmatrix} = \bar{\mathbf{x}}_{\mathbb{Z}} + \tau(\mathbf{A} \mathbf{x}) = \tau(\mathbf{y}) + \mathbf{z}_{\mathbb{Z}}. \quad (1)$$

Therefore, by setting

$$\mathbf{z} := \psi^{-1}(\mathbf{z}_{\mathbb{Z}} - \bar{\mathbf{x}}_{\mathbb{Z}}) \in \mathcal{R}^k, \quad (2)$$

we have $\mathbf{A} \mathbf{x} = \mathbf{y} + \mathbf{z} \bmod Q$. This means that $\text{Approx}.\mathbf{A}^{-1}(\cdot)$ correctly works.

Then, we show that

$$\left\{ (\mathbf{A}, \mathbf{x}, \mathbf{y}, \mathbf{z}) : \begin{array}{l} \mathbf{y} \leftarrow \mathcal{R}_Q^k, \\ \mathbf{x} \leftarrow \text{Approx}.\mathbf{A}^{-1}(\mathbf{y}) \\ \mathbf{z} := \mathbf{y} - \mathbf{A} \mathbf{x} \bmod Q \end{array} \right\} \approx_s \left\{ (\mathbf{A}, \mathbf{x}, \mathbf{y}, \mathbf{z}) : \begin{array}{l} \mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R}, \sigma_1}^{2k}, \\ \mathbf{z} \leftarrow (\mathcal{D}_{\mathcal{R}, \sigma_1} * \mathcal{R}_p)^k, \\ \mathbf{y} := \mathbf{A} \mathbf{x} + \mathbf{z} \bmod Q \end{array} \right\}. \quad (3)$$

By Theorem 2.7, we have

$$\left\{ (\mathbf{A}_{\mathbb{Z}}, \mathbf{x}_{\mathbb{Z}}, \mathbf{y}_{\mathbb{Z}}, \mathbf{z}_{\mathbb{Z}}) : \begin{array}{l} \mathbf{y}_{\mathbb{Z}} \leftarrow \mathbb{Z}_Q^n, \\ \mathbf{x}_{\mathbb{Z}} \leftarrow \text{Approx}_{\mathbb{Z}}.\mathbf{A}_{\mathbb{Z}}^{-1}(\mathbf{y}_{\mathbb{Z}}), \\ \mathbf{z}_{\mathbb{Z}} := \mathbf{y}_{\mathbb{Z}} - \mathbf{A}_{\mathbb{Z}} \mathbf{x}_{\mathbb{Z}} \bmod Q \end{array} \right\} \quad (4)$$

$$\approx_s \left\{ (\mathbf{A}_{\mathbb{Z}}, \mathbf{x}_{\mathbb{Z}}, \mathbf{y}_{\mathbb{Z}}, \mathbf{z}_{\mathbb{Z}}) : \begin{array}{l} \mathbf{x}_{\mathbb{Z}} \leftarrow \$ \mathcal{D}_{\mathbb{Z}, \sigma_1}^{3n}, \\ \mathbf{z}_{\mathbb{Z}} \leftarrow \$ \mathbb{Z}_p^n, \\ \mathbf{y}_{\mathbb{Z}} := \mathbf{A}_{\mathbb{Z}} \mathbf{x}_{\mathbb{Z}} + \mathbf{z}_{\mathbb{Z}} \bmod Q \end{array} \right\}. \quad (5)$$

By Equations (1) and (2), The distribution in Equation (3) follows directly from the distribution in Equations (4) and (5). Combining the above facts with Theorem 2.7, it holds that the distributions in Equation (3) are statistically close.

Furthermore, $\mathbf{A} = (\bar{\mathbf{A}} \| p \cdot \mathbf{I}_{\mathbb{Z}, nk} + \bar{\mathbf{A}} \mathbf{S} + \mathbf{E})$ is computationally indistinguishable from uniform random assuming the $\text{MLWE}_{k,k,Q,\sigma_0}$ assumption, since $(\bar{\mathbf{A}} \| \bar{\mathbf{A}} \mathbf{S} + \mathbf{E})$ is pseudorandom under the $\text{MLWE}_{k,k,Q,\sigma_0}$ assumption. $\square$

4 Construction of Our IBE Scheme

In this section, we present our IBE scheme Π .

4.1 Construction

For reference, we provide the parameters of Π in Table 3.

Table 3: Overview of parameters and notations used in Π .

Parameter	Explanation
(p, q, Q)	Modulus $Q = pq$
$\mathcal{R}, \mathcal{R}_Q$	Polynomial rings $\mathcal{R} = \mathbb{Z}[X]/(X^n + 1)$ and $\mathcal{R}_Q = \mathcal{R}/Q\mathcal{R}$
k	Dimension of public matrix $\mathbf{A} \in \mathcal{R}_Q^{k \times 2k}$
σ_{msk}	Gaussian parameter for the master secret key $\mathbf{T}_{\mathbf{A}}$
σ_{sk}	Gaussian parameter for secret keys $\mathbf{x}_{\text{id}}$
(σ, τ)	Gaussian parameters for encryption
$\mathcal{M}$	Message space $\mathcal{M} = \{0, 1\}^n \subset \mathcal{R}$
ℓ_{id}	Identity-length

Our IBE scheme $\Pi = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$ is given as follows. Our scheme uses a hash function H modeled as a (quantum) random oracle in the security proof. $H : \{0, 1\}^{\ell_{\text{id}}} \rightarrow \mathcal{R}_Q^k$ maps an identity $\text{id} \in \{0, 1\}^{\ell_{\text{id}}}$ to a random vector in $\mathcal{R}_Q^k$.

- $\text{Setup}(1^\lambda) \rightarrow (\text{msk}, \text{mpk})$:
 1. Sample $(\mathbf{A}, \mathbf{T}_{\mathbf{A}}) \leftarrow \$ \text{AppTrapGen}(1^k, p, q, \sigma_{\text{msk}})$.
 2. Output $(\text{msk} := \mathbf{T}_{\mathbf{A}} \in \mathcal{R}^{2k \times k}, \text{mpk} := \mathbf{A} \in \mathcal{R}_Q^{k \times 2k})$
- $\text{KGen}(\text{msk} = \mathbf{T}_{\mathbf{A}}, \text{mpk} = \mathbf{A}, \text{id} \in \{0, 1\}^{\ell_{\text{id}}}) \rightarrow \text{sk}_{\text{id}}$:
 1. Compute $\mathbf{y}_{\text{id}} := H(\text{id})$.
 2. Sample $\mathbf{x}_{\text{id}} \leftarrow \$ \text{AppSampPre}(\mathbf{A}, \mathbf{T}_{\mathbf{A}}, \mathbf{y}_{\text{id}}, \sigma_{\text{sk}})$.
 3. Output $\text{sk}_{\text{id}} := \mathbf{x}_{\text{id}} \in \mathcal{R}^{2k}$.
- $\text{Enc}(\text{mpk} = \mathbf{A}, \text{id} \in \{0, 1\}^{\ell_{\text{id}}}, \mathbf{M} \in \mathcal{M}) \rightarrow \text{ct}$:
 1. Compute $\mathbf{y}_{\text{id}} := H(\text{id})$.
 2. Sample $\mathbf{r} \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^k$, $\mathbf{e}_1 \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^{2k}$, and $e_2 \leftarrow \$ \mathcal{D}_{\mathcal{R}, \tau}$.
 3. Compute $\mathbf{c}_1^\top := \mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top \bmod Q$ and $c_2 := \mathbf{r}^\top \mathbf{y}_{\text{id}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot \mathbf{M} \bmod Q$.

4. Output $\text{ct} := (\mathbf{c}_1, c_2) \in \mathcal{R}_Q^{2k} \times \mathcal{R}_Q$.
- $\text{Dec}(\text{mpk} = \mathbf{A}, \text{sk}_{\text{id}} = \mathbf{x}_{\text{id}}, \text{ct} = (\mathbf{c}_1, c_2)) \rightarrow M'$:
 1. Output $M' := \lfloor \frac{2}{Q} \rfloor \cdot (c_2 - \mathbf{c}_1^\top \mathbf{x}_{\text{id}})$.

4.2 Correctness

Here, we show the correctness of the above IBE scheme Π . Suppose that the ciphertext $\text{ct} = (\mathbf{c}_1, c_2)$ and the secret key $\text{sk}_{\text{id}} = \mathbf{x}_{\text{id}}$ are correctly generated. When the Dec algorithm operates as specified, we have

$$\begin{aligned}
c_2 - \mathbf{c}_1^\top \mathbf{x}_{\text{id}} &= \mathbf{r}^\top \mathbf{y}_{\text{id}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M - (\mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top) \mathbf{x}_{\text{id}} \\
&= \mathbf{r}^\top \mathbf{y}_{\text{id}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M - \mathbf{r}^\top \mathbf{A} \mathbf{x}_{\text{id}} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}} \\
&= \mathbf{r}^\top \mathbf{y}_{\text{id}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M - \mathbf{r}^\top \mathbf{y}_{\text{id}} - \mathbf{r}^\top \mathbf{z} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}} \\
&= \lfloor \frac{Q}{2} \rfloor \cdot M + \underbrace{e_2 - \mathbf{r}^\top \mathbf{z} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}}}_{\text{noise}}.
\end{aligned}$$

Here, we use the fact that $\mathbf{A} \mathbf{x}_{\text{id}} = \mathbf{y}_{\text{id}} + \mathbf{z} \pmod{Q}$ holds, where $\mathbf{z} \in \mathcal{R}^k$. By Theorem 2.2, $\|\mathbf{r}\|_\infty \leq \sqrt{\lambda}\sigma$, $\|\mathbf{e}_1\|_\infty \leq \sqrt{\lambda}\sigma$, and $\|e_2\|_\infty \leq \sqrt{\lambda}\tau$ hold. In addition, by Theorem 3.5, $\|\mathbf{x}_{\text{id}}\|_\infty \leq \sqrt{\lambda}\sigma_{\text{sk}}$ and $\|\mathbf{z}\|_\infty \leq p + \sqrt{\lambda}\sigma_{\text{sk}}$. Thus, the infinity norm of noise is bounded by

$$\begin{aligned}
\|\text{noise}\|_\infty &= \|e_2 - \mathbf{r}^\top \mathbf{z} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}}\|_\infty \\
&\leq \|e_2\|_\infty + \|\mathbf{r}^\top \mathbf{z}\|_\infty + \|\mathbf{e}_1^\top \mathbf{x}_{\text{id}}\|_\infty \\
&\leq \sqrt{\lambda}\tau + nk \cdot \|\mathbf{r}\|_\infty \cdot \|\mathbf{z}\|_\infty + 2nk \cdot \|\mathbf{e}_1\|_\infty \cdot \|\mathbf{x}_{\text{id}}\|_\infty \\
&\leq \sqrt{\lambda}\tau + nk\sqrt{\lambda}\sigma(p + 3\sqrt{\lambda}\sigma_{\text{sk}}).
\end{aligned}$$

For the correctness, we need $\|\text{noise}\|_\infty \leq Q/4$. We will set the parameters below so that the upper bound is less than $Q/4$.

4.3 Asymptotic Parameters

We set the parameters of the scheme Π to satisfy the following conditions. The following conditions apply to our scheme. Under the following conditions, we provide comparisons in Table 1.

- $\epsilon = \epsilon(\lambda) > 0$ is negligible.
- AppTrapGen and AppSampPre operate properly (Theorem 3.5): That is, $Q = pq$

$$\begin{aligned}
\sigma_{\text{sk}}^2 &\geq (q^2 + 1) \cdot (3kn\lambda\sigma_{\text{msk}}^2 + 1) \cdot \ln(2nk(1 + 1/\epsilon))/\pi \\
&\geq (q^2 + 1) \cdot (3kn\lambda\sigma_{\text{msk}}^2 + 1) \cdot \eta_\epsilon(\mathcal{R}^k)^2.
\end{aligned}$$

- Correctness holds: That is, $Q/4 \geq \sqrt{\lambda}\tau + nk\sqrt{\lambda}\sigma(p + 3\sqrt{\lambda}\sigma_{\text{sk}})$.
- The $\text{MLWE}_{k,k,Q,\sigma_{\text{msk}}}$ assumption holds (Theorem 2.9): That is, $\sigma_{\text{msk}} = \sqrt{k} \cdot \omega(\sqrt{\log n})$.
- The $\text{ahMLWE}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}$ assumption holds (Theorem 3.2): That is, there exist $s, t \geq 2\sqrt{2}$ with
 - $\sigma_0 \geq \sqrt{\ln(2nk(1 + 1/\epsilon))/\pi} \geq \eta_\epsilon(\mathcal{R}^k)$ and $\tau_0 \geq \sqrt{\ln(2n(1 + 1/\epsilon))/\pi} \geq \eta_\epsilon(\mathcal{R})$,
 - $t\tau_0 \geq \frac{\sqrt{(s^2+1)(s^2+2)}}{s} \sigma_0(p + \sqrt{\lambda}\sigma_{\text{sk}})$,

- $\sigma := \sqrt{s^2 + 1}\sigma_0$, $\tau := \sqrt{t^2 + 1}\tau_0$, and $\sigma^* := s/2 \cdot \sigma_0$,
- the $\text{MLWE}_{k,2k,Q,\sigma^*}$ assumption holds, i.e., $\sigma^* = \sqrt{2k} \cdot \omega(\sqrt{\log n})$.
- Conditions for Theorem 3.3 holds: That is, n is a power of 2 and $X^n + 1$ splits into n linear factor modulo prime p , and

$$\tau, \sigma_{\text{sk}} \geq q\sqrt{n \ln(2n(3k+1)(1+1/\epsilon))}/\pi \cdot p^{\frac{k+1}{3k+1} + \frac{\epsilon}{k+1}}.$$

Candidate Asymptotic Parameters. We give a set of asymptotic parameters that fit the above conditions.³

- $\epsilon = 2^{-\Theta(\log^2 \lambda)}$.
- k is a constant such that $\frac{k+1}{3k+1} + \frac{\epsilon}{k+1} \leq \frac{1}{2}$.⁴
- $n = \Theta(\lambda)$.
- $\sigma_{\text{msk}} = \Theta(\log \lambda)$.
- $\sigma_{\text{sk}} = \Theta(\lambda^8 \log^7 \lambda)$.
- $(\sigma, \tau) = (\Theta(\log \lambda), \Theta(\lambda^{8.5} \log^8 \lambda))$.
- $(p, q) = (\Theta(\lambda^5 \log^4 \lambda), \Theta(\lambda^5 \log^4 \lambda))$.

For the above candidates, it is possible to set $k = O(1)$, $n = O(\lambda)$, and $Q = \text{poly}(\lambda)$. Table 1 shows that our scheme is asymptotically the most efficient among LWE-, RLWE-, and MLWE-based schemes.

5 Security Proof in the Random Oracle Model

In this section, we prove the following theorem.

Theorem 5.1. If the $\text{MLWE}_{k,k,Q,\sigma_{\text{msk}}}$ and $\text{ahMLWE}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}$ assumptions hold, our IBE scheme Π in Section 4.1 satisfies adaptive-identity anonymity in the random oracle model. In particular, for any classical PPT adversary $\mathcal{A}$ making at most Q_{H} random oracle queries to H and Q_{id} secret key queries, there exist two classical PPT reduction algorithms $\mathcal{B}_1$ and $\mathcal{B}_2$ such that

$$\text{Adv}_{\mathcal{A},\Pi}^{\text{IBE}}(\lambda) \leq \text{Adv}_{k,k,Q,\sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1) + \text{Adv}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}^{\text{ahMLWE}}(\lambda, \mathcal{B}_2) + \text{negl}(\lambda).$$

Proof. Let $\mathcal{A}$ be a classical PPT adversary attacking the adaptive-identity anonymity of Π . Without loss of generality, we assume that whenever $\mathcal{A}$ queries a secret key or asks for a challenge ciphertext, the corresponding id has already been queried to the random oracle H .

We show the security of Π via the following games. In each game, we define E_i as the event that $\mathcal{A}$ wins in Game_i .

Game₀ : This is the real security game. At the beginning of the game, the challenger $\mathcal{C}$ first runs $\text{Setup}(1^\lambda)$ to obtain (mpk, msk) and then gives mpk to $\mathcal{A}$. $\mathcal{C}$ then samples $\text{coin} \leftarrow \{0, 1\}$ and keeps it secret. During the game, $\mathcal{A}$ can make many random oracle and key generation queries and one challenge query. For each query, $\mathcal{C}$ behaves as follows:

- When $\mathcal{A}$ makes a random oracle query to H on id , $\mathcal{C}$ samples a random polynomial $\mathbf{y}_{\text{id}} \leftarrow \mathcal{R}_Q^k$ and locally stores the tuple $(\text{id}, \mathbf{y}_{\text{id}}, \perp)$, and returns $\mathbf{y}_{\text{id}}$ to $\mathcal{A}$.

³Although our scheme allows for flexible parameter settings, we provide examples of parameters that achieve compactness asymptotically equivalent to the NTRU-based scheme.

⁴As long as $k \geq 2$ and $\epsilon \leq 3/14$, this condition is always satisfied.

- When $\mathcal{A}$ makes a key generation query for id such that $(\text{id}, *) \notin \mathcal{Q}_{\text{id}}$, $\mathcal{C}$ returns $\text{sk}_{\text{id}} := \mathbf{x}_{\text{id}}$, where $\mathbf{x}_{\text{id}} \leftarrow \$ \text{AppSampPre}(\mathbf{A}, \mathbf{T}_{\mathbf{A}}, \mathbf{y}_{\text{id}}, \sigma_{\text{sk}})$.
- When $\mathcal{A}$ makes the challenge query for the challenge identity id^* and a message $\mathbf{M}^*$, $\mathcal{C}$ returns $\text{ct}^* = (c_1^*, c_2^*) \leftarrow \$ \text{Enc}(\text{mpk}, \text{id}^*, \mathbf{M}^*)$ if $\text{coin} = 0$ and $\text{ct}^* \leftarrow \$ \mathcal{R}_Q^{2k+1}$ if $\text{coin} = 1$.

At the end of the game, $\mathcal{A}$ outputs a guess $\widehat{\text{coin}}$ for coin . Finally, $\mathcal{C}$ outputs $\widehat{\text{coin}}$.

By definition, we have

$$\left| \Pr[\mathbf{E}_0] - \frac{1}{2} \right| = \left| \Pr[\widehat{\text{coin}} = \text{coin}] - \frac{1}{2} \right| = \text{Adv}_{\mathcal{A}, \Pi}^{\text{IBE}}(\lambda).$$

Game₁ : This is the same as **Game₀** except how $\mathcal{C}$ answers the random oracle queries. Upon $\mathcal{A}$'s random oracle query on id in **Game₁**, $\mathcal{C}$ first samples $\mathbf{x}_{\text{id}} \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma_{\text{sk}}}^{2k}$ and $\mathbf{z}_{\text{id}} \leftarrow \$ (\mathcal{D}_{\mathcal{R}, \sigma_{\text{sk}}} * \mathcal{R}_p)^k$ and sets $\mathbf{y}_{\text{id}} := \mathbf{A}\mathbf{x}_{\text{id}} + \mathbf{z}_{\text{id}} \bmod Q$. Then, $\mathcal{C}$ locally stores $(\text{id}, \mathbf{y}_{\text{id}}, (\mathbf{x}_{\text{id}}, \mathbf{z}_{\text{id}}))$ and returns $\mathbf{y}_{\text{id}}$.

Based on our choice of parameters, we can apply Theorem 3.5, which ensures that all $\mathbf{y}_{\text{id}}$ are statistically close to uniform as in **Game₀**. Thus, the statistical distance between the view of $\mathcal{A}$ in **Game₀** and **Game₁** is $Q_H \cdot \text{negl}(\lambda) = \text{negl}(\lambda)$. Therefore, we have

$$|\Pr[\mathbf{E}_0] - \Pr[\mathbf{E}_1]| = \text{negl}(\lambda).$$

Game₂ : This is the same as **Game₁** except how $\mathcal{C}$ generates secret keys $\mathbf{x}_{\text{id}}$. In particular, $\mathcal{C}$ does not use the trapdoor $\mathbf{T}_{\mathbf{A}}$ to generate them. When $\mathcal{C}$ generates $\mathbf{x}_{\text{id}}$ for id , $\mathcal{C}$ does not run the **AppSampPre** algorithm but retrieves the unique tuple $(\text{id}, \mathbf{y}_{\text{id}}, (\mathbf{x}_{\text{id}}, \mathbf{z}_{\text{id}}))$ from local storage and returns $\text{sk}_{\text{id}} := \mathbf{x}_{\text{id}}$.

Based on our choice of parameters, we can apply Theorem 3.5, which ensures that $\mathbf{x}$ in **Game₁** sampled by the **AppSampPre** algorithms distribute statistically close to $\mathcal{D}_{\mathcal{R}, \sigma_{\text{sk}}}^{2k}$ conditioned on $\mathbf{y}_{\text{id}}$. Since $\mathcal{A}$ obtains at most Q_{id} secret keys, we have

$$|\Pr[\mathbf{E}_1] - \Pr[\mathbf{E}_2]| = Q_{\text{id}} \cdot \text{negl}(\lambda) = \text{negl}(\lambda).$$

Game₃ : This is the same as **Game₂** except how $\mathcal{C}$ generates a master public key $\mathbf{A}$. In **Game₃**, $\mathcal{C}$ does not run the **AppTrapGen** algorithm but samples a uniformly random matrix $\mathbf{A} \leftarrow \$ \mathcal{R}_Q^{k \times 2k}$. Since $\mathcal{C}$ did not use a master secret key $\mathbf{T}_{\mathbf{A}}$ to answer $\mathcal{A}$'s queries in **Game₂**, it can answer all $\mathcal{A}$'s queries.

By Theorem 3.5, the $\text{MLWE}_{k, k, Q, \sigma_{\text{msk}}}$ assumption ensures that **Game₂** and **Game₃** are computationally indistinguishable. Then, there exists a PPT algorithm $\mathcal{B}_1$ such that

$$|\Pr[\mathbf{E}_2] - \Pr[\mathbf{E}_3]| = \text{Adv}_{k, k, Q, \sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1).$$

Game₄ : This game is the same as **Game₃** except how $\mathcal{C}$ generates a challenge ciphertext ct^* when $\text{coin} = 0$.

In **Game₃**, $\mathcal{C}$ samples $\mathbf{r} \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^k$, $\mathbf{e}_1 \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^{2k}$, and $e_2 \leftarrow \$ \mathcal{D}_{\mathcal{R}, \tau}$, and computes

$$\mathbf{c}_1^{*\top} := \mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top \bmod Q, \quad c_2^* := \mathbf{r}^\top \mathbf{y}_{\text{id}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot \mathbf{M} \bmod Q.$$

In **Game₄**, $\mathcal{C}$ first retrieves the unique tuple $(\text{id}^*, \mathbf{y}_{\text{id}^*}, (\mathbf{x}_{\text{id}^*}, \mathbf{z}_{\text{id}^*}))$ from local storage. Then, $\mathcal{C}$ samples $\mathbf{r} \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^k$, $\mathbf{e}_1 \leftarrow \$ \mathcal{D}_{\mathcal{R}, \sigma}^{2k}$, and $e_2 \leftarrow \$ \mathcal{D}_{\mathcal{R}, \tau}$, and computes

$$\mathbf{c}_1^{*\top} := \mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top \bmod Q, \quad c_2^* := \boxed{\mathbf{c}_1^{*\top} \mathbf{x}_{\text{id}^*} + \mathbf{r}^\top \mathbf{z}_{\text{id}^*} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}^*}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot \mathbf{M} \bmod Q.$$

This change is conceptual because

$$\begin{aligned}
c_2^* &= \mathbf{c}_1^{*\top} \mathbf{x}_{\text{id}^*} + \mathbf{r}^\top \mathbf{z}_{\text{id}^*} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}^*} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M \\
&= \mathbf{r}^\top \mathbf{A} \mathbf{x}_{\text{id}^*} + \mathbf{e}_1^\top \mathbf{x}_{\text{id}^*} + \mathbf{r}^\top \mathbf{z}_{\text{id}^*} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}^*} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M \\
&= \mathbf{r}^\top (\mathbf{y}_{\text{id}^*} - \mathbf{z}_{\text{id}^*}) + \mathbf{r}^\top \mathbf{z}_{\text{id}^*} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M \\
&= \mathbf{r}^\top \mathbf{y}_{\text{id}^*} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M.
\end{aligned}$$

Therefore, we have

$$\Pr[\mathbf{E}_3] = \Pr[\mathbf{E}_4].$$

Game₅ : This is the same as **Game₄** except how $\mathcal{C}$ generates $\mathbf{c}_1^*$ when $\text{coin} = 0$. In **Game₅**, $\mathcal{C}$ computes $\mathbf{c}_1^* := \mathbf{c} + \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 \bmod Q$ instead of $\mathbf{c}_1^* := \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 \bmod Q$, where $\mathbf{c} \leftarrow \mathcal{R}_Q^{2k}$.

The $\text{ahMLWE}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}$ assumption ensures that **Game₄** and **Game₅** are computationally indistinguishable. To show this, we use $\mathcal{A}$ to construct an ahMLWE adversary $\mathcal{B}_2$ as follows:

1. $\mathcal{B}_2$ is given $\mathbf{A} \in \mathcal{R}_Q^{k \times 2k}$ from the ahMLWE challenger $\mathcal{C}_{\text{ahMLWE}}$ and sends $\text{mpk} := \mathbf{A}$ to $\mathcal{A}$.
2. $\mathcal{B}_2$ answers $\mathcal{A}$'s random oracle and key generation queries as in **Game₄**.
3. Upon $\mathcal{A}$'s challenge query on $(\text{id}^*, \mathbf{M}^*)$, $\mathcal{B}_2$ retrieves the tuple $(\text{id}^*, \mathbf{y}_{\text{id}^*}, (\mathbf{x}_{\text{id}^*}, \mathbf{z}_{\text{id}^*}))$ from local storage, and sends $\mathbf{z} := \begin{pmatrix} \mathbf{z}_{\text{id}^*} \\ -\mathbf{x}_{\text{id}^*} \end{pmatrix}$ to $\mathcal{C}_{\text{ahMLWE}}$. Note that, it holds that $\|\mathbf{z}\|_\infty \leq \beta$.
4. $\mathcal{B}_2$ gives $(\mathbf{u} := \mathbf{c} + \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 \bmod Q, h := \mathbf{r}^\top \mathbf{z}_{\text{id}^*} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}^*} + e')$ from $\mathcal{C}_{\text{ahMLWE}}$, where $\mathbf{c} = \mathbf{0}$ or $\mathbf{c} \leftarrow \mathcal{R}_Q^{2k}$, and $e' \leftarrow \mathcal{D}_{\mathcal{R},\tau}$. Then, $\mathcal{B}_2$ sets

$$\mathbf{c}_1^* := \mathbf{u}, \quad c_2^* := \mathbf{u}^\top \mathbf{x}_{\text{id}^*} + h + \lfloor \frac{Q}{2} \rfloor \cdot \mathbf{M}^* \bmod Q,$$

and send $(\mathbf{c}_1^*, c_2^*)$ as the challenge ciphertext to $\mathcal{A}$.

5. $\mathcal{B}_2$ receives $\widehat{\text{coin}}$ from $\mathcal{A}$, it outputs $\widehat{\text{coin}}$.

If $\mathbf{c} = \mathbf{0}$, $\mathbf{c}_1^*$ follows the same distribution as in **Game₄**. Otherwise, $\mathbf{c}_1^*$ follows the same distribution as in **Game₅**. Thus, we complete the reduction, and we have

$$|\Pr[\mathbf{E}_4] - \Pr[\mathbf{E}_5]| = \text{Adv}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}^{\text{ahMLWE}}(\lambda, \mathcal{B}_2).$$

Game₆ : This game is the same as **Game₅** except how $\mathcal{C}$ computes ct^* when $\text{coin} = 0$. In **Game₆**, $\mathcal{C}$ computes

$$\mathbf{c}_1^{*\top} := \mathbf{c}^\top + \mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top \bmod Q, \quad c_2^* := \boxed{\mathbf{c}^\top \mathbf{x}_{\text{id}^*} + \mathbf{r}^\top \mathbf{y}_{\text{id}^*}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M \bmod Q.$$

This change is conceptual. Therefore, we have

$$\Pr[\mathbf{E}_5] = \Pr[\mathbf{E}_6].$$

Game₇ : This is the same as **Game₆** except how $\mathcal{C}$ generates c_2^* . Regardless of the value coin , $\mathcal{C}$ samples $c_2^* \leftarrow \mathcal{R}_Q$.

We show that **Game₆** and **Game₇** are statistically indistinguishable. Based on our choice of parameters, we can apply Theorem 3.3, which ensures that $\mathbf{c}^{*\top} \mathbf{x}_{\text{id}^*} + e_2 \bmod Q$ is statistically close to uniform even given $\mathbf{y}_{\text{id}^*} = \mathbf{A} \mathbf{x}_{\text{id}^*} + \mathbf{z}_{\text{id}^*} \bmod Q$. Therefore, the statistical distance between the view of $\mathcal{A}$ in **Game₆** and **Game₇** is $\text{negl}(\lambda)$ and we have

$$|\Pr[\mathbf{E}_6] - \Pr[\mathbf{E}_7]| = \text{negl}(\lambda).$$

Game₈ : This is the same as **Game₇** how $\mathcal{C}$ generates $\mathbf{c}_1^*$. Regardless of the value coin , $\mathcal{C}$ samples $\mathbf{c}_1^* \leftarrow \mathcal{R}_Q^{2k}$. Thus, we have

$$\Pr[\mathbf{E}_8] = \frac{1}{2}.$$

Since this change does not affect the view of $\mathcal{A}$ at all, then we have

$$\Pr[\mathbf{E}_7] = \Pr[\mathbf{E}_8].$$

By combining everything, we have

$$\text{Adv}_{\mathcal{A}, \Pi}^{\text{IBE}}(\lambda) \leq \text{Adv}_{k, k, Q, \sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1) + \text{Adv}_{k, 2k, Q, \sigma, \tau, p + \sqrt{\lambda}\sigma_{\text{sk}}}^{\text{ahMLWE}}(\lambda, \mathcal{B}_2) + \text{negl}(\lambda).$$

□

By combining Theorem 5.1 and Theorem 3.2, we obtain the following corollary.

Corollary 5.2. Let $\sigma^* \in \mathbb{R}$ be a positive real number that satisfies the conditions of Theorem 2.6 for $\beta = p + \sqrt{\lambda}\sigma_{\text{sk}}$. For any classical PPT adversary $\mathcal{A}$ making at most Q_{H} random oracle queries to H and Q_{id} secret key queries, there exist two classical PPT reduction algorithms $\mathcal{B}_1$ and $\mathcal{B}_2$ such that

$$\text{Adv}_{\mathcal{A}, \Pi}^{\text{IBE}}(\lambda) \leq \text{Adv}_{k, k, Q, \sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1) + \text{Adv}_{k, 2k, Q, \sigma^*}^{\text{MLWE}}(\lambda, \mathcal{B}_2) + \text{negl}(\lambda).$$

6 Security Proof in the Quantum Random Oracle Model

This section provides the security proof of our scheme in the quantum random oracle model (QROM). To do this, we recall the foundations of the QROM with reference to [KYY21, Tak21]. We refer to [NC10] for more details.

6.1 Preliminaries on the QROM

Quantum Computation. Let $|0\rangle := (1, 0)^\top$ and $|1\rangle := (0, 1)^\top$ denote the state of 1 qubit. Let $|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle \in \mathbb{C}^{2^n}$ denote the state of n qubits, where $\alpha_x \in \mathbb{C}$ satisfying $\sum_{x \in \{0,1\}^n} |\alpha_x|^2 = 1$ and $|x\rangle = |x_1 x_2 \dots x_n\rangle = |x_1\rangle \otimes \dots \otimes |x_n\rangle$ for $x_1, \dots, x_n \in \{0,1\}$ is an orthonormal basis on $\mathbb{C}^{2^n}$ called the computational basis. If we measure the state $|\psi\rangle$ in the computational basis, the classical bit $x \in \{0,1\}^n$ is observed with probability $|\alpha_x|^2$ and the state becomes $|x\rangle$.

An arbitrary evolution of quantum state from $|\psi\rangle$ to $|\psi'\rangle$ is described by a unitary matrix $\mathbf{U}$, where $|\psi'\rangle = \mathbf{U}|\psi\rangle$. In short, a quantum algorithm is described by quantum evolutions that consist of evolutions with unitary matrices and measurements. The running time of a quantum algorithm $\mathcal{A}$ is defined as the number of universal gates and measurements required to execute $\mathcal{A}$. If $\mathcal{A}$ is a quantum oracle algorithm, we assume that $\mathcal{A}$ runs in a unit of time. Any efficient classical computation can be achieved efficiently by quantum computation. In particular, for any function f that is classically computable, there exists a unitary matrix $\mathbf{U}_f$ such that $\mathbf{U}_f|x, y\rangle = |x, f(x) \oplus y\rangle$, and the number of universal gates to express $\mathbf{U}_f$ is linear in the size of a classical circuit that computes f .

QROM. The notion of the QROM was introduced by Boneh et al. [BDF⁺11] as an extension of the (classical) random oracle model (ROM) in a quantum world. In the case of the ROM, the QROM is an idealized model, where a hash function is idealized to be an oracle that simulates a random function. On the other hand, as opposed to the ROM, the hash function in the QROM is a quantumly accessible oracle. In security proofs in the QROM, a random function $\text{H} : \mathbf{X} \rightarrow \mathbf{Y}$ is uniformly chosen at the beginning, and an adversary can make queries on a quantum state $\sum_{x,y} \alpha_{x,y} |x\rangle |y\rangle$ to the oracle and receive $\sum_{x,y} \alpha_{x,y} |x\rangle |\text{H}(x) \oplus y\rangle$.

Let $\mathcal{A}^{|\text{H}\rangle}$ denote a quantum algorithm that can quantumly access the oracle $|\text{H}\rangle$. As shown in [Zha12], quantum random oracles can be simulated by a family of $2Q_{\text{H}}$ -wise independent hash functions for an adversary that quantumly accesses the random oracle at most Q_{H} times.

Lemma 6.1 ([Zha12]). Any quantum algorithm $\mathcal{A}$ making quantum queries to random oracles can be efficiently simulated by a quantum algorithm $\mathcal{B}$, which has the same output distribution but makes no queries.

6.2 Security Proof in the QROM

Theorem 6.2. If the $\text{MLWE}_{k,k,Q,\sigma_{\text{msk}}}$ and $\text{ahMLWE}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}$ assumptions hold, our IBE scheme Π in Section 4.1 achieves adaptively anonymous in the quantum random oracle model. In particular, for any quantum PPT adversary $\mathcal{A}$ making at most Q_H random oracle queries to H and Q_{id} secret key queries, there exist two quantum polynomial time reduction algorithms $\mathcal{B}_1$ and $\mathcal{B}_2$ such that

$$\text{Adv}_{\mathcal{A},\Pi}^{\text{IBE}}(\lambda) \leq \text{Adv}_{k,k,Q,\sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1) + \text{Adv}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}^{\text{ahMLWE}}(\lambda, \mathcal{B}_2) + \text{negl}(\lambda).$$

Proof. We show the security of Π via the following games. In each game, we define E_i as the event that $\mathcal{A}$ wins in Game_i . Let $\text{Samp}(\sigma_{\text{sk}}, p; r)$ be a PPT algorithm that, given a Gaussian parameter σ_{sk} , a positive integer p , and a random coin $r \in \{0,1\}^{\ell_r}$, outputs $(\mathbf{x}, \mathbf{z})$, where $\mathbf{x}$ sampled from a distribution statistically close to $\mathcal{D}_{\mathcal{R},\sigma_{\text{sk}}}^{2k}$ and $\mathbf{z}$ sampled from a distribution statistically close to $(\mathcal{D}_{\mathcal{R},\sigma_{\text{sk}}} * \mathcal{R}_p)^k$.

Game₀ : This is the actual security game. At the beginning of the game, the challenge $\mathcal{C}$ chooses a random function $H : \{0,1\}^{\ell_{\text{id}}} \rightarrow \mathcal{R}_Q^k$. Then, it generates $(\text{msk}, \text{mpk}) \leftarrow \text{Setup}(1^\lambda)$ and gives mpk to the adversary $\mathcal{A}$. Then, it samples $\text{coin} \leftarrow \{0,1\}$ and keeps it secret. During the game, $\mathcal{A}$ can make many (quantum) random oracle and secret key queries and one challenge query. These queries are handled as follows:

- When $\mathcal{A}$ makes a (quantum) random oracle query on a quantum state $\sum_{\text{id},y} \alpha_{\text{id},y} |\text{id}\rangle |y\rangle$, $\mathcal{C}$ returns $\sum_{\text{id},y} \alpha_{\text{id},y} |\text{id}\rangle |H(\text{id}) \oplus y\rangle$.
- When $\mathcal{A}$ makes a key generation query for id such that $(\text{id}, *) \notin Q_{\text{id}}$, $\mathcal{C}$ returns $\text{sk}_{\text{id}} := \mathbf{x}_{\text{id}}$, where $\mathbf{x}_{\text{id}} \leftarrow \text{AppSampPre}(\mathbf{A}, \mathbf{T}_{\mathbf{A}}, \mathbf{y}_{\text{id}}, \sigma_{\text{sk}})$.
- When $\mathcal{A}$ makes a challenge query (id^*, M^*) , $\mathcal{C}$ returns $\text{ct}^* \leftarrow \text{Enc}(\text{mpk}, \text{id}^*, M^*)$ if $\text{coin} = 0$ and $\text{ct}^* \leftarrow \mathcal{R}_Q^{2k+1}$ if $\text{coin} = 1$.

At the end of the game, $\mathcal{A}$ outputs a guess $\widehat{\text{coin}}$ for coin . Finally, $\mathcal{C}$ outputs $\widehat{\text{coin}}$.

By definition, we have

$$\left| \Pr[E_0] - \frac{1}{2} \right| = \left| \Pr[\widehat{\text{coin}} = \text{coin}] - \frac{1}{2} \right| = \text{Adv}_{\mathcal{A},\Pi}^{\text{IBE}}(\lambda).$$

Game₁ : This is the same as **Game₀** except how $\mathcal{C}$ answers the quantum random oracle queries. First, $\mathcal{C}$ picks a $2Q_H$ -wise independent hash function $h_{2Q_H} : \{0,1\}^{\ell_{\text{id}}} \rightarrow \{0,1\}^{\ell_r}$. Then, we define $H(\text{id}) := \mathbf{A}\mathbf{x}_{\text{id}} + \mathbf{z}_{\text{id}} \bmod Q$, where $(\mathbf{x}_{\text{id}}, \mathbf{z}_{\text{id}}) := \text{Samp}(\sigma_{\text{sk}}, p; h_{2Q_H}(\text{id}))$ and use this H throughout the game.

For any fixed id , the distribution of $H(\text{id})$ is identical, and its statistical distance from the uniform distribution is $\text{negl}(\lambda)$ due to Theorem 3.5. Note that in this game, we only change the distribution of $\mathbf{y}_{\text{id}}$ for each identity, and how we create secret keys is unchanged. Then, due to Theorem 6.1, we have

$$|\Pr[E_0] - \Pr[E_1]| = \text{negl}(\lambda).$$

Game₂ : This is the same as **Game₁** except how $\mathcal{C}$ generates secret keys $\mathbf{x}_{\text{id}}$. By the end of this game, $\mathcal{C}$ will no longer require the trapdoor $\mathbf{T}_{\mathbf{A}}$ to generate the secret keys. When $\mathcal{A}$ queries a secret key for id , $\mathcal{C}$ returns $\text{sk}_{\text{id}} := \mathbf{x}_{\text{id}}$, where $(\mathbf{x}_{\text{id}}, \mathbf{z}_{\text{id}}) := \text{Samp}(\sigma_{\text{sk}}, p; h_{2Q_H}(\text{id}))$.

By following the same argument in **Game₂** of the proof of Theorem 5.1, we have

$$|\Pr[E_1] - \Pr[E_2]| = Q_{\text{id}} \cdot \text{negl}(\lambda) = \text{negl}(\lambda).$$

Game₃ : This is the same as **Game₂** except how $\mathcal{C}$ generates a master public key $\mathbf{A}$. In **Game₃**, $\mathcal{C}$ does not run the **AppTrapGen** algorithm but samples a uniformly random matrix $\mathbf{A} \leftarrow \mathcal{R}_Q^{k \times 2k}$.

By following the same argument in **Game₃** of the proof of Theorem 5.1, we have

$$|\Pr[\mathbf{E}_2] - \Pr[\mathbf{E}_3]| = \text{Adv}_{k,k,Q,\sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1).$$

Game₄ : This game is the same as **Game₃** except how $\mathcal{C}$ generates a challenge ciphertext ct^* when $\text{coin} = 0$. In **Game₃**, $\mathcal{C}$ samples $\mathbf{r} \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^k$, $\mathbf{e}_1 \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^{2k}$, and $e_2 \leftarrow \mathcal{D}_{\mathcal{R},\tau}$, and computes

$$\mathbf{c}_1^{*\top} := \mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top \bmod Q, \quad c_2^* := \mathbf{r}^\top \mathbf{y}_{\text{id}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M \bmod Q.$$

In **Game₄**, $\mathcal{C}$ first retrieves the unique tuple $(\text{id}^*, \mathbf{y}_{\text{id}^*}, (\mathbf{x}_{\text{id}^*}, \mathbf{z}_{\text{id}^*}))$ from local storage. Then, $\mathcal{C}$ samples $\mathbf{r} \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^k$, $\mathbf{e}_1 \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^{2k}$, and $e_2 \leftarrow \mathcal{D}_{\mathcal{R},\tau}$, and computes

$$\mathbf{c}_1^{*\top} := \mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top \bmod Q, \\ c_2^* := \mathbf{c}_1^{*\top} \mathbf{x}_{\text{id}^*} + \mathbf{r}^\top \mathbf{z}_{\text{id}^*} - \mathbf{e}_1^\top \mathbf{x}_{\text{id}^*} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M \bmod Q.$$

By following the same argument in **Game₄** of the proof of Theorem 5.1, we have

$$\Pr[\mathbf{E}_3] = \Pr[\mathbf{E}_4].$$

Game₅ : This is the same as **Game₄** except how $\mathcal{C}$ generates $\mathbf{c}_1^*$ when $\text{coin} = 0$. In **Game₅**, $\mathcal{C}$ computes $\mathbf{c}_1^* := \mathbf{c} + \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 \bmod Q$ instead of $\mathbf{c}_1^* := \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 \bmod Q$, where $\mathbf{c} \leftarrow \mathcal{R}_Q^{2k}$.

By following the same argument in **Game₅** of the proof of Theorem 5.1, we have

$$|\Pr[\mathbf{E}_4] - \Pr[\mathbf{E}_5]| = \text{Adv}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}^{\text{ahMLWE}}(\lambda, \mathcal{B}_2).$$

Game₆ : This game is the same as **Game₅** except how $\mathcal{C}$ computes ct^* when $\text{coin} = 0$. In **Game₆**, $\mathcal{C}$ computes

$$\mathbf{c}_1^{*\top} := \mathbf{c}^\top + \mathbf{r}^\top \mathbf{A} + \mathbf{e}_1^\top \bmod Q, \quad c_2^* := \mathbf{c}^\top \mathbf{x}_{\text{id}^*} + \mathbf{r}^\top \mathbf{y}_{\text{id}} + e_2 + \lfloor \frac{Q}{2} \rfloor \cdot M \bmod Q.$$

By following the same argument in **Game₆** of the proof of Theorem 5.1, we have

$$\Pr[\mathbf{E}_5] = \Pr[\mathbf{E}_6].$$

Game₇ : This is the same as **Game₆** except how $\mathcal{C}$ generates c_2^* . Regardless of the value coin , $\mathcal{C}$ samples $c_2^* \leftarrow \mathcal{R}_Q$.

By following the same argument in **Game₇** of the proof of Theorem 5.1, we have

$$|\Pr[\mathbf{E}_6] - \Pr[\mathbf{E}_7]| = \text{negl}(\lambda).$$

Game₈ : This is the same as **Game₇** how $\mathcal{C}$ generates $\mathbf{c}_1^*$. Regardless of the value coin , $\mathcal{C}$ samples $\mathbf{c}_1^* \leftarrow \mathcal{R}_Q^{2k}$.

By following the same argument in **Game₈** of the proof of Theorem 5.1, we have Thus, we have

$$\Pr[\mathbf{E}_7] = \Pr[\mathbf{E}_8], \quad \Pr[\mathbf{E}_8] = \frac{1}{2}.$$

Therefore, by combining everything, we have

$$\text{Adv}_{\mathcal{A},\Pi}^{\text{IBE}}(\lambda) \leq \text{Adv}_{k,k,Q,\sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1) + \text{Adv}_{k,2k,Q,\sigma,\tau,p+\sqrt{\lambda}\sigma_{\text{sk}}}^{\text{ahMLWE}}(\lambda, \mathcal{B}_2) + \text{negl}(\lambda).$$

□

By combining Theorem 6.2 and Theorem 3.2, we obtain the following corollary.

Corollary 6.3. Let $\sigma^* \in \mathbb{R}$ be a positive real number that satisfies the conditions of Theorem 2.6 for $\beta = p + \sqrt{\lambda}\sigma_{\text{sk}}$. For any quantum PPT adversary $\mathcal{A}$ making at most Q_{H} random oracle queries to H and Q_{id} secret key queries, there exist two quantum PPT reduction algorithms $\mathcal{B}_1$ and $\mathcal{B}_2$ such that

$$\text{Adv}_{\mathcal{A}, \Pi}^{\text{IBE}}(\lambda) \leq \text{Adv}_{k, k, Q, \sigma_{\text{msk}}}^{\text{MLWE}}(\lambda, \mathcal{B}_1) + \text{Adv}_{k, 2k, Q, \sigma^*}^{\text{MLWE}}(\lambda, \mathcal{B}_2) + \text{negl}(\lambda).$$

7 Conclusion

In this paper, we proposed a new compact and tightly secure (anonymous) IBE scheme based on the MLWE assumption, known as the standard lattice assumption, in the QROM. Our IBE scheme is the first to achieve the asymptotically most compact and tight security under the standard lattice assumption. In addition to the aforementioned academic contributions, it is important from a practical standpoint to construct an efficient IBE scheme based on MLWE. This is because it shares the same assumptions as ML-KEM, the U.S. standard for post-quantum KEM/PKE, making it well-suited for real-world implementation. Since our focus is on asymptotic efficiency, we will leave further optimization and implementation evaluations for future work.

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